

Research paper

D-ECOMposer: Sustainable part decomposition for additive manufacturing using machine learning based life cycle assessment

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ABSTRACT

Additive Manufacturing (AM) has garnered significant attention due to its potential for sustainable production. To further enhance this potential, Design for Additive Manufacturing (DfAM) methodologies are frequently employed. However, traditional design approaches often fall short in addressing the inherent limitations of AM, such as build size constraints, extended lead time, and the necessity for support structure. However, due to these limitations, part decomposition (PD) has recently gained prominence as a viable solution. While the benefits of PD might be less pronounced if a model can be produced in its entirety on a single AM device, this study assumes scenarios where the model is too large for the build space of the AM device, making decomposition necessary. This study proposes a grid-based PD method that utilizes machine learning-based Life Cycle Assessment (LCA) to minimize environmental impact. The experimental data in this study were collected and analyzed based on the FDM (Fused deposition modeling) process. Initially, a predictive model is developed to quickly and accurately estimate the carbon footprint of a design candidate based on the geometric characteristics of a 3D model. This predictive model is subsequently employed as the objective function in the optimization of PD using a genetic algorithm (GA). To validate the efficacy of the proposed method, experiments were conducted on four test models using FDM. While this study focuses on FDM, the proposed methodology has potential applicability to other AM processes. The experimental results clearly demonstrate that the proposed method outperforms traditional empirical approaches in reducing the carbon footprint.

1. Introduction

Sustainability is increasingly recognized as a critical issue for the future, drawing significant attention to technologies that address this challenge. Among these, Additive Manufacturing (AM) is particularly recognized as an effective manufacturing method that can drive a sustainable future [1]. AM excels in producing complex shapes by building materials layer by layer. This process can be further advanced through Design for Additive Manufacturing (DfAM). DfAM methodologies, such as topology optimization, lattice structures, part consolidation (PC), and part decomposition (PD), offer significant advantages in terms of cost and energy consumption. For instance, GE Aviation redesigned a helicopter engine, originally composed of approximately 900 parts, into 16 parts using DfAM, achieving material savings and supply chain simplification [2]. Thus, AM offers numerous opportunities for sustainability, leveraging its high design freedom.

Numerous efforts have been made to demonstrate the sustainability of AM. Many studies have focused on redesigning existing products

using DfAM and quantitatively assessing the environmental impact. Particularly, previous studies have aimed at reducing material consumption and manufacturing time through PC, thereby decreasing environmental impact. However, these studies faced scalability limitations due to inherent constraints of AM, such as build size restrictions, prolonged manufacturing time, and the necessity for support structure. Recently, PD has emerged as a strategy to overcome these limitations. PD involves decomposing a single product into multiple subcomponents [3], allowing for several advantages: it enables production within build size constraints [4], minimizes support structure usage, reduces build time by adjusting part orientation and placement [5], and addresses quality issues such as the staircase effect inherent to AM [6]. However, PD introduces a trade-off between decomposition and assembly. This paper focuses on the sustainability aspect of PD for the first time. We explore the potential to maximize AM sustainability by decomposing to minimize the carbon footprint. Additionally, we propose the use of common modules to enhance recyclability, a key concept of

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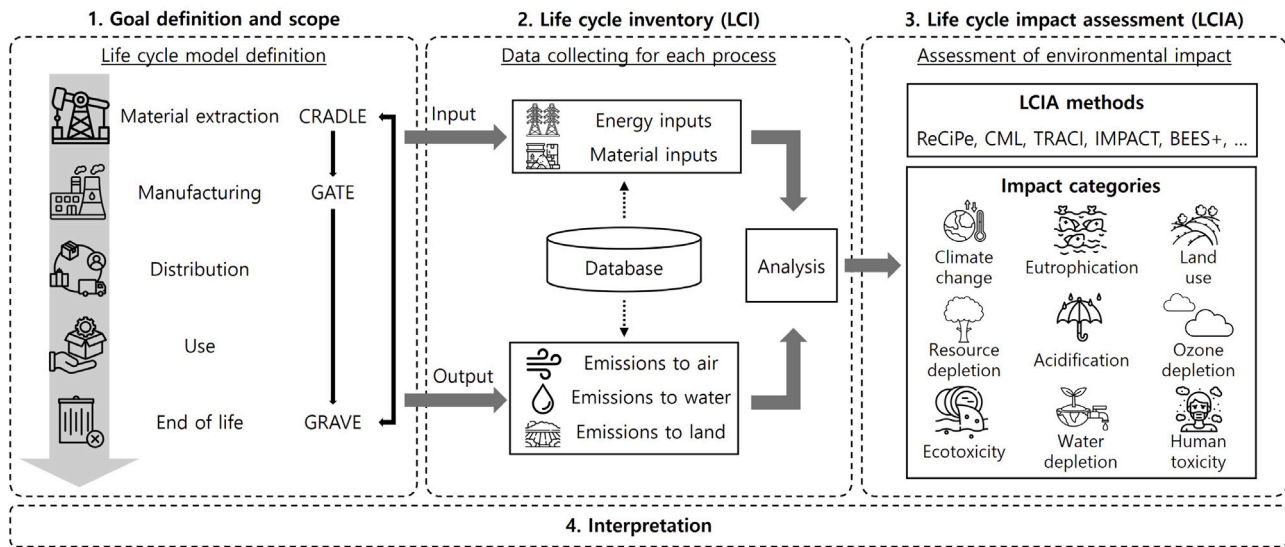


Fig. 1. Life cycle assessment process consisting of 4 stages based on ISO 14040.

the circular economy, aiming to extend product life cycle and minimize resource waste. Common modules, standardized components reusable across various products, can enhance manufacturing efficiency and reduce waste. This study presents a PD framework, D-EComposer, that seeks to minimize the carbon footprint while maximizing recyclability through the use of common modules.

Life Cycle Assessment (LCA) is a tool used to quantitatively evaluate the environmental impact of each stage of a product's life cycle. Consequently, LCA has been employed to quantitatively analyze the sustainability of AM. However, traditional LCA methodologies present challenges due to their high barriers to entry and lengthy processing time, limiting accessibility [7]. This has led to a demand for improved methods and tools that facilitate easier environmental impact assessment. This paper proposes an automated environmental impact assessment method using a machine learning-based model that leverages design information. The proposed machine learning-based environmental impact prediction model is utilized as an objective function in the optimization of PD, aiming to enhance sustainability.

The key contributions of this paper are summarized as follows:

- A sustainable PD method is proposed utilizing common modules to enhance recyclability.
- A machine learning model is developed capable of predicting environmental impact using only design information, supporting sustainability-focused decision-making in the design phase.
- The proposed machine learning-based environmental impact prediction model is used as an objective function in PD optimization to maximize sustainability.

This paper is structured as follows: Section 2 reviews the existing literature. Section 3 provides an overview of the proposed D-EComposer framework. D-EComposer is a comprehensive methodology designed to derive optimal PD strategies that minimize the carbon footprint during the early stages of design. Section 4 introduces the machine learning-based environmental impact prediction model and explains how it is utilized as a core element of D-EComposer. Section 5 proposes an optimization framework for grid-based PD and presents experimental results to analyze the effectiveness of D-EComposer. Finally, Section 6 concludes the study by summarizing the findings and discussing potential areas for further research, such as the application of the D-EComposer framework to different industries.

2. Related work

2.1. LCA in AM

Assessing the environmental impact of manufacturing processes is crucial for evaluating sustainability. AM has recently garnered significant attention as a sustainable manufacturing method, and quantitatively evaluating its environmental impact throughout the entire manufacturing process is necessary to substantiate this claim. Such analysis is essential for meeting global environmental certification standards, such as ISO 14001 [8]. LCA, as standardized by ISO 14044 [9], has been widely employed to quantitatively evaluate environmental impacts.

According to ISO 14040 [10], LCA consists of four stages, as shown in Fig. 1. In the first stage, the scope and specific objectives of the assessment within the life cycle are defined. The assessment can consider the range from raw material extraction to the manufacturing stage (Cradle to Gate), from raw material extraction to disposal (Cradle to Grave), or even include the recycling process (Cradle to Cradle). In the second stage, the life cycle inventory (LCI) is analyzed, wherein all processes within the selected life cycle scope are identified and quantified as inputs (materials and energy) and outputs (emissions). In the third stage, an appropriate method for evaluating the selected environmental impacts is chosen, known as life cycle impact assessment (LCIA), with methods such as ReCiPe, CML, and TRACI being commonly used. In the final stage, the impacts identified in the LCIA are assessed to understand their environmental implications.

Several studies have compared the overall environmental impacts of AM and conventional manufacturing (CM) [11–13] or conducted comparisons among various AM processes [14,15]. In addition to these comparisons, recent studies have explored the sustainability benefits of AM across different industries. For instance, in the automotive sector, lightweight designs produced through AM have demonstrated reduced material usage and lower energy consumption during the use phase, aligning with decarbonization goals [16]. Similarly, in the biomedical field, the production of Ti-6Al-4V knee implants using electron beam melting (EBM) resulted in reduced material waste compared to conventional machining, enhancing overall sustainability [17]. Powder bed fusion (PBF) has also been highlighted for its environmental advantages in producing orthopedic devices, as it minimizes waste and supports material efficiency when compared to subtractive manufacturing [18]. Beyond specific applications, studies have also analyzed AM's sustainability potential across multiple industries. For instance, additive

manufacturing combined with foam materials has demonstrated significant resource efficiency and lightweighting benefits in the mechanical and medical engineering fields, as shown through laser beam melting and open-cell metal structures [19]. Furthermore, a comprehensive review by Hegab et al. [20] emphasized that AM technologies are applied in diverse sectors, including food, medical, automotive, and aerospace, highlighting their ability to reduce CO₂ emissions and material waste while improving energy efficiency. Huang et al. [12] analyzed the sustainability benefits of lightweighting when transitioning the production of aircraft components from CM to AM. This study highlighted the potential for significant greenhouse gas reductions within a Cradle to Gate scope by adopting AM. On the other hand, Yoon et al. [21] warned that energy consumption in AM could be up to 100 times higher than in traditional bulk forming processes, which could result in lower sustainability performance. Watson and Taminger [22] evaluated that while AM is more energy-efficient for small products, machining may be more efficient for larger products. Furthermore, research in the broader AM field has also addressed integrating sustainability and resource prediction into early design stages, particularly through methodologies tailored for LPBF (Laser powder bed fusion) processes, which highlight the potential of predictive frameworks to advance eco-friendly manufacturing [23–25]. These findings suggest that a complex interplay of factors must be considered to ensure the environmental sustainability of manufacturing processes.

2.2. AI-based LCA

Despite the potential of LCA tools, many engineers find it challenging to effectively use them due to their complexity and the time-intensive nature of traditional LCA methods. In most cases, LCA is applied after product development is complete [26,27], making it difficult to incorporate LCA results into early design decisions [28]. Thus, there is a need to introduce LCA-based environmental impact assessments early in product development to support sustainability-focused decision-making. However, methods that enable quick and easy prediction of environmental impacts at the design stage are still lacking.

Given these challenges, artificial intelligence (AI) techniques, such as machine learning and deep learning, are increasingly being used to predict various datasets. Among these, artificial neural network (ANN), which mimic human neurons, is particularly well-suited for solving complex prediction problems based on datasets with nonlinear relationships [29]. ANN has been applied in LCA as well. For instance, D'Amico et al. [30] proposed an ANN-based LCA framework for predicting the environmental impact of buildings at the design stage. In the chemical industry, Hou et al. [31] proposed a predictive model to estimate the quantity of pollutants generated throughout the life cycle.

While AI applications in LCA research have been explored across various industries, their use in predicting the environmental impacts of AM remains relatively underdeveloped. Only a limited number of studies have applied AI techniques in this context. Notably, Naser et al. [32] introduced a framework designed to automate the LCA process for AM through the development of data-driven predictive models. In their study, data was extracted from STereoLithography (STL) and G-Code files generated during the design and manufacturing stages, and nine geometry-related parameters along with five material-related parameters closely associated with environmental impacts were selected for analysis. This methodology aimed to streamline the quantitative evaluation of AM's environmental impact. In subsequent research, Naser et al. [33] advanced this work by developing a machine learning-based LCA framework that considers the interaction between AM processes and product-process co-design. Their approach involved creating a data-driven LCA predictive model that accurately reflects the interactions between various AM processes and product designs, achieving high prediction accuracy through machine learning algorithms. This work has provided a more efficient and precise method for predicting the environmental impact of AM.

2.3. DfAM

DfAM is a methodology aimed at achieving various benefits such as weight reduction and assembly time reduction through the (re)design of parts. Based on the changes in the number of parts before and after (re)design, studies can be classified into those that reduce the number of parts (part consolidation, PC), increase the number of parts (part decomposition, PD), or optimize the shape while maintaining the number of parts [34].

To date, much research has focused on PC, where multiple parts are merged into a single component. PC has primarily been used to achieve weight reduction. For example, Yang et al. [35] reduced a triple clamp assembly from 19 parts to 7 parts, achieving a 20% weight reduction. Additionally, some studies have compared the environmental impact before and after PC using LCA. Yang et al. [36] investigated the environmental impact differences between assembly design and post-PC parts manufactured through AM. Their findings suggested that AM could be more environmentally friendly if the weight of a throttle pedal is reduced by over 30%. More recently, Son et al. [37] demonstrated that PC could significantly improve sustainability throughout the product life cycle.

Topology optimization has also been highlighted as a key design tool for improving environmental sustainability in AM. Prathyusha et al. [38] reviewed how integrating topology optimization with AM enables lightweight design, reducing material usage and costs, which in turn mitigates the environmental impact in the automotive and aerospace sectors. Munk et al. [39] demonstrated that topology optimization of aircraft components can significantly enhance lifecycle sustainability by extending fatigue life and reducing maintenance time. Additionally, Hoschke et al. [25] proposed a sustainability-oriented design framework by integrating LCA models into AM-based topology optimization processes, enabling early-stage ecological footprint estimation and its incorporation into design. On the other hand, Borda et al. [40] highlighted that topology optimization of automotive components, when considering the characteristics of manufacturing processes throughout their lifecycle, can be the most environmentally friendly. They emphasized that AM, with its ability to fabricate complex shapes using minimal material, can maximize the potential for sustainability. Lastly, Kumar et al. [41] showed that AM-based topology optimization for customized orthotic and assistive devices in the medical field can reduce ecological footprints while simultaneously improving patient satisfaction.

On the other hand, PD has been increasingly utilized to overcome the inherent limitations of AM. PD research can be divided into studies proposing evaluation criteria and those developing decomposition methodologies. Luo et al. [42] proposed a framework called Chopper for PD, which formalizes decomposition criteria considering factors such as assemblability and structural integrity. Yao et al. [43] focused on overcoming AM limitations by proposing PD with goals of fast production and space-saving. Jiang et al. [4] introduced a PD method aimed at ensuring the structural integrity of decomposed parts. Oh et al. [34] classified existing PD research from the perspectives of printability, productivity, functionality, artistry, and interoperability. Demir et al. [44] proposed a near-convex segmentation that minimizes an energy function considering four component properties. Inspired by their study, Oh et al. [3] proposed eight evaluation criteria for PD, considering manufacturability and assemblability, by referring to ISO/ASTM 52910. Along with this, various studies have been conducted on methods for performing part decomposition. Chen et al. [45, 46] proposed a pyramid-based decomposition technique to maximize the self-supporting capability of parts, demonstrating that it effectively minimizes the need for support structures, especially for parts with significant overhangs. Oh et al. [5] introduced a binary space partitioning (BSP) approach, which analyzes the external and internal structures of parts to determine the optimal decomposition path, thereby improving manufacturability and assemblability. Additionally, Demir et al. [44]

explored near-convex segmentation, a technique that divides parts into multiple convex subparts to maintain joint mechanical strength while enhancing manufacturability and assemblability.

Previous PD studies have often focused on minimizing support structures and manufacturing time, which can offer sustainability benefits through reduced material and energy consumption. However, unlike PC and topology optimization studies, few PD studies have directly quantified environmental impacts using LCA or explicitly considered sustainability as a primary objective.

2.4. Eco-friendly module identification

To consider environmental impacts during the product design stage, various evaluation and development methods are employed. One of the most common methods is LCA, which assesses the environmental impact across product life cycle, thereby helping to formulate strategies to minimize the impact during the design process. However, significant challenges arise in identifying and evaluating the amount of material used and the manufacturing methods in the early design stages, as these factors heavily influence the environmental impact assessment. Additionally, the lack of standardized environmental impact assessment methods for AM further complicates this process.

Research on modular design for eco-friendly product development can be broadly categorized into two main approaches. The first approach involves conducting LCA on each module and subsequently developing eco-friendly products by minimizing the environmental impact of the required module combinations for product assembly [47]. When direct LCA evaluation of equipment is challenging, the electricity consumption during equipment operation is assessed as an indirect measure of environmental impact [48]. The second approach focuses on designing modules that support the reuse or recycling of components and materials at the end of the product life cycle [49]. This approach aims to reduce waste during the product's end-of-life phase, extend the product life cycle, and uncover environmental and economic value. Thus, LCA is essential for eco-friendly product development, and its effective utilization during the design phase is a critical issue.

When designing modules with LCA considerations, the goals are multi-faceted. Typically, the aim is to increase the complexity within the module while reducing the complexity of connections between modules. However, additional objectives must be considered when designing eco-friendly modules. These objectives include improving the recyclability and reusability of modules by considering material similarity and expected lifespan similarity between components [49], as well as reducing costs associated with the supply chain and (re) manufacturing processes [47,48,50,51]. Modules can also be designed for ease of disassembly to facilitate the reuse and recycling of components at the end of the product life cycle [52]. Additionally, modular product family design strategies are employed to achieve product diversity and enable mass production [48]. In this strategy, common modules that are used across similar products and variable modules that change according to specific requirements are developed. The core of this strategy is the use of reusable common modules, which can reduce cost and time in mass production. Furthermore, the interchangeability of these common modules at the end of the product life cycle enhances reusability, contributing to eco-friendly product development [47,48,53].

In eco-friendly module design, structural and functional connectivity between components is crucial, closely aligning with the process of PC in AM. During PC, the structural and functional connectivity of the product is a primary consideration, and components clustered within the module are often targeted for consolidation [35,54]. The degree of modularization can also be adjusted by determining the number of modules within a product [55], utilizing both PC and PD techniques. While modular design principles have been widely applied in PC research, the integration of PD into modular design has been relatively unexplored [46]. Moreover, most modular design efforts

AS-IS: Manufacture the entire model



TO-BE: Manufacture only custom modules

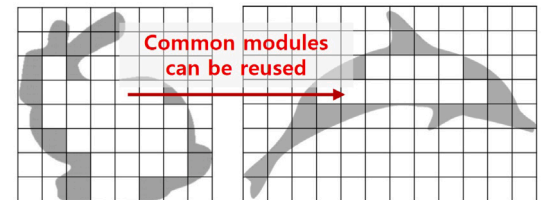


Fig. 2. The concept of manufacturing custom modules while reusing common modules after grid-based decomposition.

have focused on the component level, where a module is defined as a collection of multiple components. However, there is a growing need for research and development that defines the design scope within a single component, overcomes AM manufacturing constraints, and considers eco-friendliness in production.

3. Research framework overview

In this section, we propose a PD methodology aimed at maximizing sustainability when utilizing AM. Previous studies have primarily focused on optimizing methodologies to maximize technical goals such as manufacturability in AM. These studies concentrated on how to decompose parts and how to optimize these decompositions to enhance manufacturability. However, such approaches often did not sufficiently consider sustainability.

Recently, efforts to more accurately evaluate the environmental impact of AM and improve its sustainability have increased. In particular, studies analyzing the environmental impact of AM, such as the carbon footprint generated during the manufacturing process, have highlighted the importance of assessing these impacts and incorporating them into decision-making.

Naser et al. [32,33] responded to this need by proposing a machine learning-based LCA framework to predict the environmental impact of AM. This research developed a model that predicts environmental impacts by utilizing various design information and process parameters, emphasizing the importance of considering sustainability in the manufacturing process. However, this study had limitations in that it relied on manufacturing information as input to the predictive model, making it difficult to quickly assess environmental impacts during the early design stage with minimal design information. Additionally, the study did not practically apply the developed predictive model to AM process optimization for experimentation and validation.

To overcome these limitations, this study proposes the D-EComposer framework, which integrates a PD strategy to minimize the environmental impact of the AM process. D-EComposer develops an ANN-based LCA tool capable of predicting environmental impacts with high accuracy using minimal design information during the early design stage. In the early design phase, detailed information about manufacturing processes, material production, and supply chain data is often unavailable, making analytical approaches difficult to apply. ANN overcomes this limitation by using accessible data such as the geometric

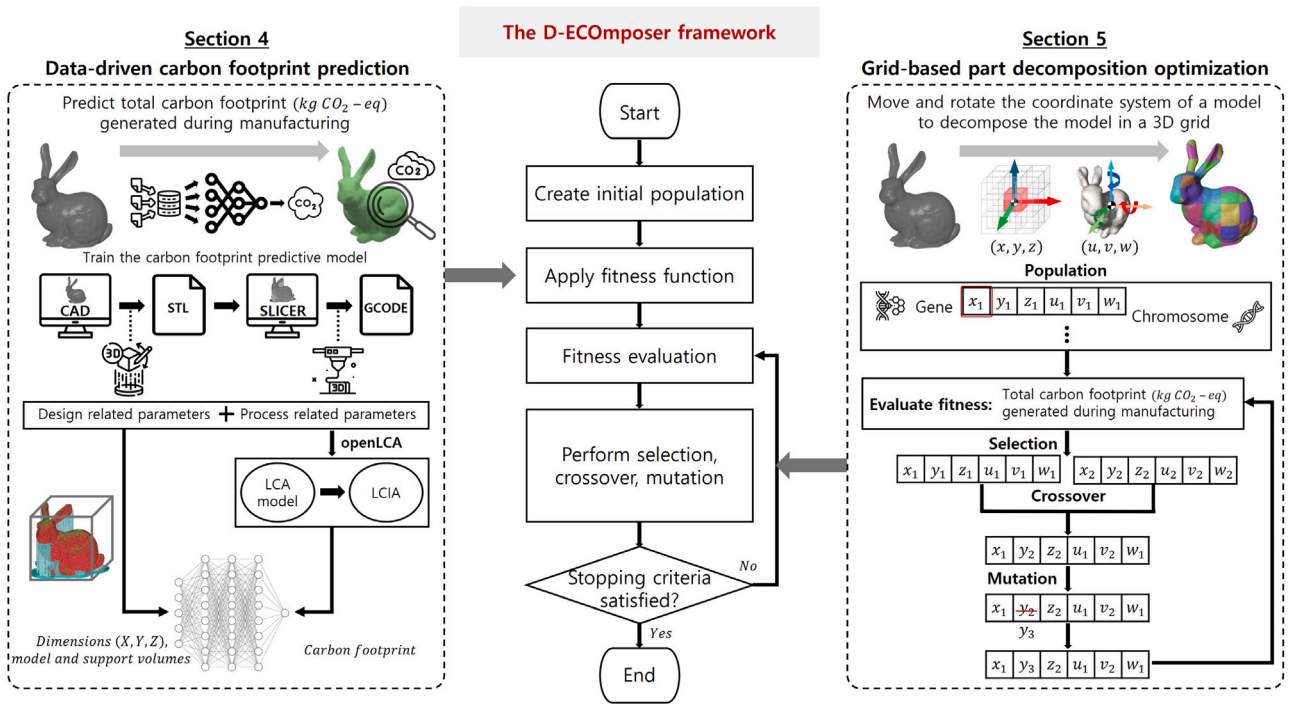


Fig. 3. The overview of the D-ECOMposer framework.

information of 3D models to rapidly predict environmental impacts. This enables immediate feedback on design changes and plays a crucial role in incorporating sustainability into design decision-making. Furthermore, during the PD optimization process, this predictive model is used as an objective function to propose a PD methodology that minimizes the carbon footprint. While the benefits of PD might be less pronounced if a model can be produced in its entirety on a single AM device, this study assumes scenarios where the model exceeds the build size of the AM device, making decomposition necessary. By doing so, the D-ECOMposer framework minimizes the environmental impact of the AM process and proposes a PD optimization strategy that considers recyclability. In this study, grid-based decomposition is applied to create common modules with good compatibility and cubic shapes that can be reused as much as possible. For example, as illustrated in Fig. 2, while the entire model would have previously been manufactured (AS-IS), the goal of the D-ECOMposer is to manufacture only custom modules by reusing common modules (TO-BE).

The feasibility of producing and reusing common modules has been demonstrated in various scenarios. For instance, Chen et al. [46] explored a modular decomposition strategy for 3D printing, which allows for the assembly and reuse of standardized modules across different designs. This approach not only reduces manufacturing complexity but also enhances the adaptability of designs for diverse applications. Similarly, Choi et al. [56] implemented a grid-based modular decomposition in the manufacturing of large-scale marine components, showcasing its effectiveness in optimizing production time and minimizing environmental impact. These examples highlight how modular design strategies, including those based on grid structures, can be practically applied to balance efficiency, adaptability, and sustainability in manufacturing processes.

To perform grid-based decomposition, it is necessary to determine where to position the model within the grid and at what angle to rotate it. Fig. 3 shows the overall flow of the methodology proposed in this study. First, a genetic algorithm (GA) is employed to perform the decomposition optimization. To conduct the optimization, it is crucial to define an objective function. In this study, the objective function is defined as the carbon footprint generated during the manufacturing of the decomposed modules, with the goal of minimizing it. For this

purpose, the carbon footprint must be quickly calculated during the optimization process. Section 4 introduces the process of training and testing a predictive model for estimating the carbon footprint as part of data-driven LCA. Section 5 explains the implementation of a GA-based grid decomposition method to obtain the optimal solution that satisfies the objective function. During this process, two additional objective functions (maximizing common modules and minimizing support structures), which are expected to contribute to sustainability, are also set as comparison groups, and the results are compared.

4. Data-driven carbon footprint prediction

To minimize the carbon footprint generated during the manufacturing of decomposed modules, it is essential to quantitatively evaluate the carbon footprint. Therefore, we needed an automated LCA process to calculate the carbon footprint of the decomposition results generated during optimization. It was particularly important to create a predictive model that estimates the carbon footprint during manufacturing based on the geometric information of a 3D model. To develop the predictive model, it is necessary to prepare a suitable dataset that can predict the carbon footprint in the AM process. For this, we first prepared design information of various 3D models suitable for AM and data on material and energy consumption during manufacturing. We then prepared carbon footprint data calculated using an LCA software based on material and energy usage. This data is used to train the predictive model, which supports accurate predictions of the carbon footprint under various design conditions. This section explains the process of creating a machine learning model for predicting carbon footprint during manufacturing.

4.1. Data preparation

In the design and manufacturing stages of AM, the primary data used includes the geometry information contained in STL files and the process planning information in G-Code files. STL files provide the geometric information of 3D models. Since the geometric characteristics of a 3D model can influence material and time consumption,

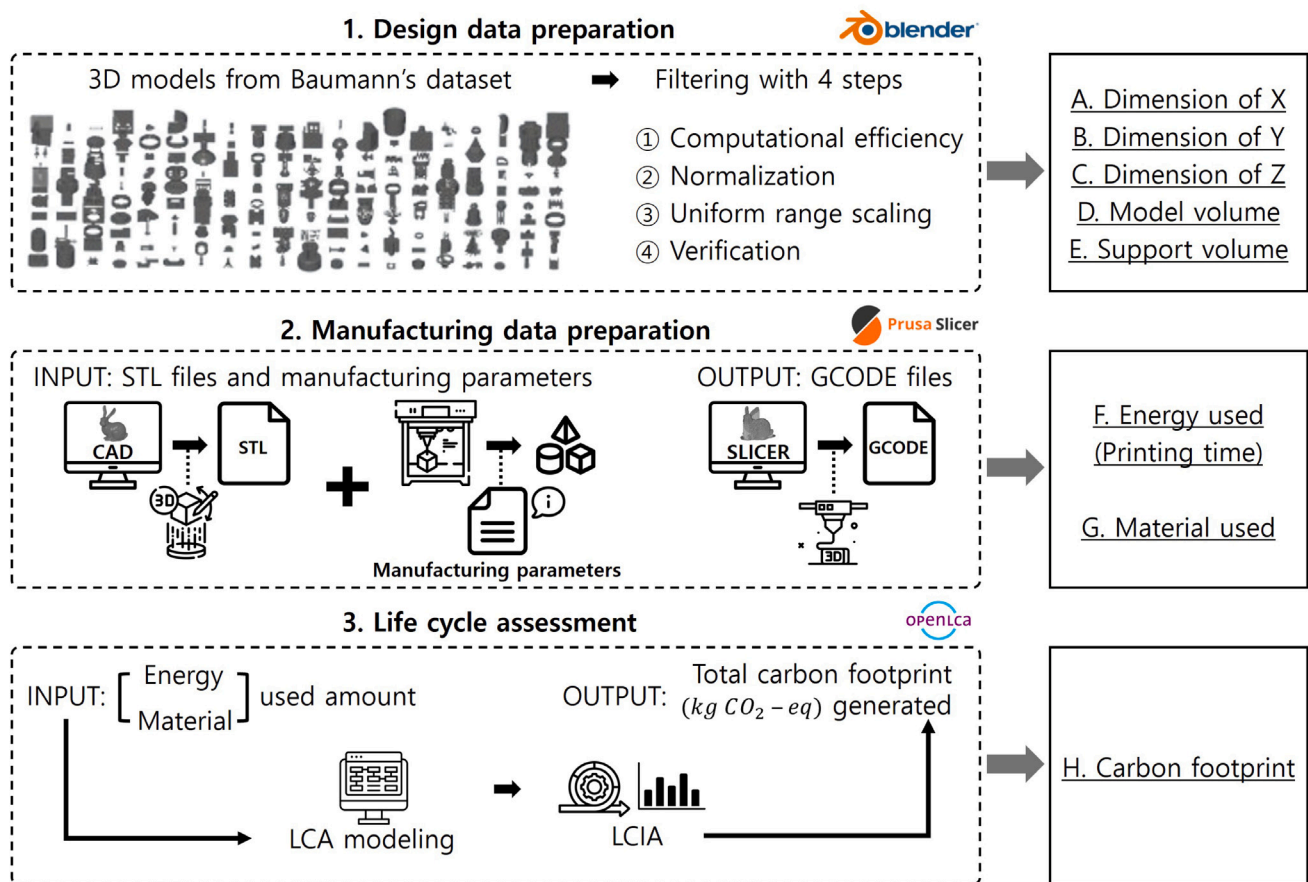


Fig. 4. Data preparation process for machine learning model training.

we assumed that these features could be used to predict the carbon footprint. The G-Code files, generated by slicing software, provide more direct information such as material and time usage during manufacturing. In this study, we used Prusa Slicer¹ to automatically extract G-Code files for all STL files. The material and time usage information obtained from each G-Code file was then used to calculate the carbon footprint with the open-source LCA tool, openLCA.² We adopted the dataset by Baumann [57], which includes 30,000 models collected from Thingiverse,³ for training. Fig. 4 shows the entire data collection process explained so far. We will now detail each step of the process.

4.1.1. Design data preparation

To efficiently extract data from 3D models in the dataset and improve the performance of the predictive model, we performed filtering on the 3D models. First, we excluded large files over 10 MB to prevent memory issues and reduce computation time during the data processing and training phases. This threshold was determined based on the hardware constraints and the complexity of the models. Each model was then normalized so that its maximum dimension would be 1 mm. The normalized models were then randomly scaled to sizes between 30 mm and 90 mm. Additionally, models with extremely elongated or thin shapes were excluded because such geometries often exceed the build volume of AM machines or pose physical stability challenges during the manufacturing process. Furthermore, these models could introduce issues of reliability or accuracy in subsequent G-code generation processes. Afterward, models with shapes that posed difficulties for reliable feature extraction were excluded upon detailed review.

To extract geometric feature data from the filtered 3D models, we used Blender⁴ to extract the width (Dimension X), length (Dimension Y), height (Dimension Z), and volume of bounding boxes. Additionally, since support structure inevitably occurs in AM, information on the support volume was also needed. In AM, support structure is generated for surfaces with angles smaller than the critical angle (usually at least 45 degrees) relative to the build platform. In this study, support structure was generated in Blender for surfaces with angles smaller than 45 degrees, as shown in Fig. 5, and their volume was calculated. This approach follows the default support generation strategies used in widely adopted slicing software such as Cura and PrusaSlicer, ensuring consistency with practical manufacturing conditions. While alternative support structures (tree, lattice) may affect material usage to some extent, the overall trend in support generation remains consistent. Therefore, the chosen method does not significantly impact the predictive model's validity. Thus, at this stage, the extracted data included each 3D model's bounding box width (Dimension X), length (Dimension Y), height (Dimension Z), model volume, and support volume, which are used as input data for the predictive model.

Unlike Prusa Slicer, which generates support structures during the build preparation stage, Blender was chosen to pre-generate support structures and accurately measure their volume. Prusa Slicer does not provide a direct quantification of support volume, making Blender more suitable for generating precise data required for predictive model training. To further improve the accuracy of support volume data, Tweaker [58] was employed to optimize the orientation of parts before support generation. This optimization minimized the required support volume, reflecting the most efficient build orientation and ensuring

¹ <https://github.com/prusa3d/PrusaSlicer>.

² <https://www.openlca.org/>.

³ <https://www.thingiverse.com/>.

⁴ <https://www.blender.org/>.

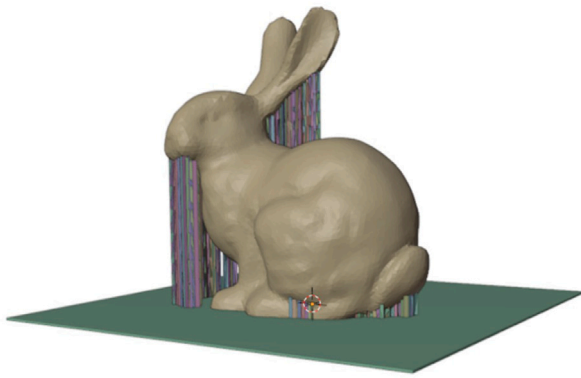


Fig. 5. An example of support structure generated in Blender. The angle threshold was set to 45 degree.

Table 1
Process parameters for generating G-Code files.

Parameter	Value
Machine type	Fused Deposition Modeling (FDM)
Material type	Acrylonitrile Butadiene Styrene (ABS)
Infill rate	100%
Average print speed	50.0 mm/s
Device power	0.5 kW

that the predictive model's input data closely aligned with practical manufacturing scenarios.

4.1.2. Manufacturing data preparation

This subsection explains the process of acquiring the necessary data for material and energy usage during manufacturing, which is essential for LCA. Before AM, slicing software is used to prepare a process plan from a 3D model. The slicing software divides the 3D model into layers and generates G-Code that instructs the AM machine to build the model layer by layer. Users specify process parameters such as layer thickness, infill ratio, and printing speed, which are then incorporated into the G-Code as commands for layer-by-layer movements, nozzle temperature, speed, and material feed rate. The process parameters used for G-Code generation in this study are listed in Table 1. Material and energy usage data, which serve as inputs for LCA, are either directly calculated by the slicing software and embedded in the G-Code or can be estimated based on the tool path described in the G-Code. In this study, we used the material usage and build time information explicitly specified in the G-Code, and we calculated energy usage based on the build time and the power consumption of the AM machine. Layer height, a critical process parameter in AM, significantly impacts build time and energy consumption. But in this study, layer height was fixed to ensure consistency across different decomposition scenarios. Fixing this parameter allowed for a clear comparison of the effects of various objective functions on decomposition results, focusing on the sustainability metrics of each strategy.

To support the predictive model and LCA calculations, Blender and PrusaSlicer were used in complementary roles. Blender was employed to pre-generate support structures and extract input data, such as support volume, while PrusaSlicer generated G-code and provided output data, including build time and material usage. This combined approach ensured accurate and comprehensive data preparation, aligning input and output parameters with real-world manufacturing conditions.

4.1.3. Life cycle assessment

To calculate the carbon footprint based on material and energy usage, we utilized the openLCA. Since this study focused on measuring the environmental impact up to the manufacturing stage, we adopted

the scope of Cradle to Gate. Fig. 6 illustrates the three-stage LCA modeling process based on the use of ABS filament and electricity during manufacturing. After LCA modeling, we constructed the LCI data, which allowed us to calculate the material and energy inputs and outputs for each process in the product life cycle. Subsequently, one of the LCIA methodologies was selected to calculate the environmental impact of interest. In this paper, the TRACI 2.1 methodology was used to calculate the carbon footprint at each stage. The following section outlines the process of constructing LCI data and performing LCIA.

(1) Raw material production: In this stage, the environmental impact of producing raw materials is calculated. Since we are targeting the Cradle to Gate scope, it is essential to consider the environmental impact from the raw material production process. Thus, we calculated the environmental impact based on the information from the Ecoinvent version 3.8 database [59], which provides environmental impact data for producing 1 kg of ABS raw material.

(2) Filament production: In this stage, the environmental impact of processing the raw material into filament is calculated. We used the energy consumption data for producing 1 kg of filament from Caceres-Mendoza et al. [60]. Similar to the raw material production stage, we calculated the environmental impact of energy consumption based on the electricity use phase data provided by the Ecoinvent database.

(3) Manufacturing: In this stage, the environmental impact generated during the AM process using filament is calculated. Since we know the energy consumption during the manufacturing stage, we again used the Ecoinvent database to calculate the environmental impact.

The environmental impact calculated at each stage is then converted into a Global Warming Potential (GWP) score using the TRACI 2.1 methodology. The GWP score is calculated by converting various greenhouse gases such as CO₂, CH₄, and N₂O into CO₂ equivalents (CO₂e) using characterization factors.

4.2. ANN architecture

Fig. 7 shows the architecture of the ANN used to predict carbon emissions in this study. The input layer receives five geometric features (Dimension X, Y, Z, part volume, and support volume). The hidden layers consist of two layers, and since there is only one prediction, the output layer has a single neuron. The ReLU activation function was used.

In this study, the AutoKeras⁵ library was used to automatically tune the hyperparameters of the ANN. AutoKeras allowed us to automatically search for the optimal model for the defined problem, saving time on model structure design and hyperparameter tuning.

4.3. Training and testing

This section presents the experimental results and analysis of the proposed predictive model's performance. The predictive model was implemented in Python 3.11.3, using TensorFlow 2.13.0 and Keras 2.13.1.

4.3.1. Model training

The input data consist of five feature types extracted from 14,000 STL files from Baumann's dataset. The output data is the carbon emission directly calculated using openLCA. The 14,000 input data was split into training and testing sets in an 8:2 ratio. For validation, 20% of the training set was randomly selected and shuffled at each training epoch. The Adam optimizer [61] was used during training. The loss function used during training was the root mean square error (RMSE).

⁵ <https://autokeras.com/>.

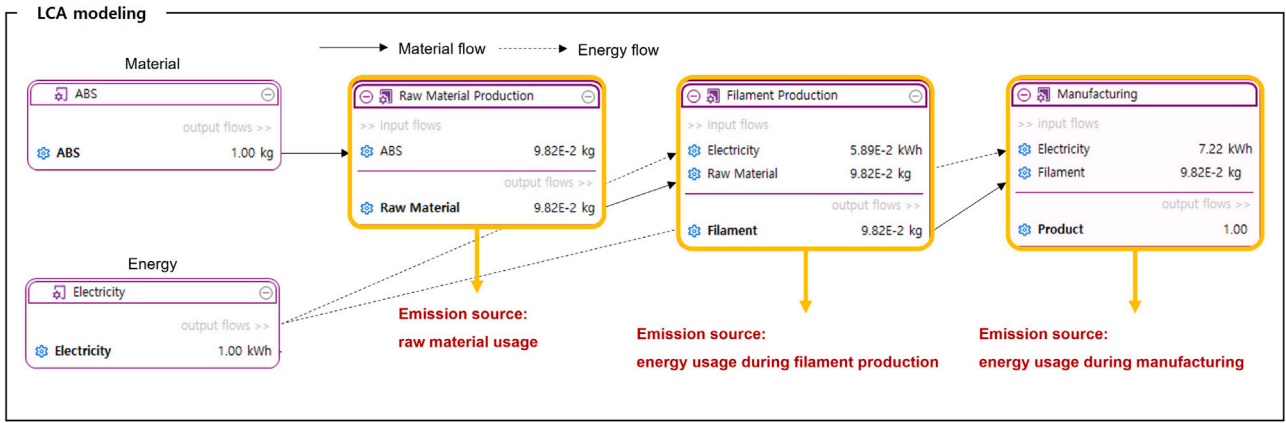


Fig. 6. LCA modeling in openLCA from Cradle to Gate.

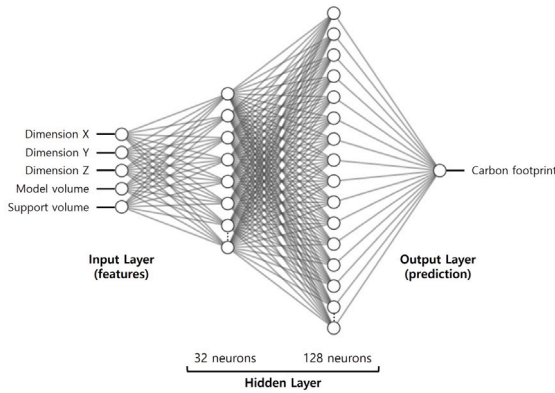


Fig. 7. ANN model architecture.

RMSE penalizes larger errors more heavily, making it advantageous for reducing large errors. It is defined as follows:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (1)$$

where:

- n is the number of data points,
- y_i is the actual value,
- $\hat{y}_i$ is the predicted value.

Fig. 8(a) shows the training curve of the predictive model. If the training curve is below the validation curve with a significant deviation, it indicates overfitting, where the model relies heavily on the training data, leading to higher errors in the test data. Conversely, if the training curve is significantly above the validation curve, it indicates underfitting, suggesting the dataset is too simple. The figure shows that the model was well-trained without overfitting or underfitting.

4.3.2. Model testing

Once training is complete, predictions are made using the test data. Fig. 8(b) and (c) show the relationship between the actual carbon emission calculated using openLCA and the predicted values generated by the model. In Fig. 8(b), the distribution of error values between the two is shown, with most error values close to zero. In Fig. 8(c), the blue data points align with the diagonal, indicating minimal deviation for most cases. While some data points show deviations of 20%–40% or more, these do not significantly impact the ANN model's primary purpose of capturing relative trends among design options. This capability enables

Table 2

Comparison of 5-fold cross-validation results for different models.

Model	Average R^2	95% Confidence interval
ANN	0.9266 ± 0.0235	[0.9038, 0.9494]
Linear Regression	0.9321 ± 0.0169	[0.9156, 0.9486]
Random Forest	0.9035 ± 0.0210	[0.8825, 0.9245]
XGBoost	0.7883 ± 0.0165	[0.7723, 0.8043]
GBM	0.7974 ± 0.0227	[0.7747, 0.8201]
LightGBM	0.8113 ± 0.0153	[0.7960, 0.8266]
SVR	0.9070 ± 0.0127	[0.8944, 0.9196]

designers to compare the sustainability of different alternatives and make informed decisions. The results confirm that the predictions were made with high reliability, as reflected by the R-squared value, which has been added to Fig. 8(c)'s caption.

The performance of the model was evaluated using the R^2 score (coefficient of determination), which is widely used for assessing regression models. The R^2 score is defined as:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (2)$$

where:

- y_i is the actual value,
- $\hat{y}_i$ is the predicted value,
- $\bar{y}$ is the mean of the actual values.

The R^2 score ranges from 0 to 1, with a score closer to 1 indicating that the model's predictions fit the data well. To further validate the robustness of the model, a 5-fold cross-validation was conducted, and the performance of the ANN model was benchmarked against other regression models. The results are summarized in Table 2. Table 2 shows that the ANN model achieved an average R^2 of 0.9266 ± 0.0235 with a 95% confidence interval of [0.9038, 0.9494]. The ANN model performed comparably to Linear Regression, which had an average R^2 of 0.9321 ± 0.0169 , and outperformed Random Forest, XGBoost, GBM, LightGBM and SVR under certain conditions. Notably, ANN showed stronger performance in folds with high support volume involvement, where non-linear relationships were more significant. This highlights the ANN model's capability to learn intricate multivariate relationships. The cross-validation results confirm the ANN model's robustness and adaptability. Furthermore, the flexibility of the ANN architecture positions it well for integrating new variables in the future, such as support density, joint area, or other manufacturing-related factors. These characteristics make ANN a strong candidate for accurately capturing the complex dynamics of additive manufacturing processes.

Fig. 9 shows the results of the Monte Carlo simulation conducted to quantify uncertainty in LCA predictions. The green line represents

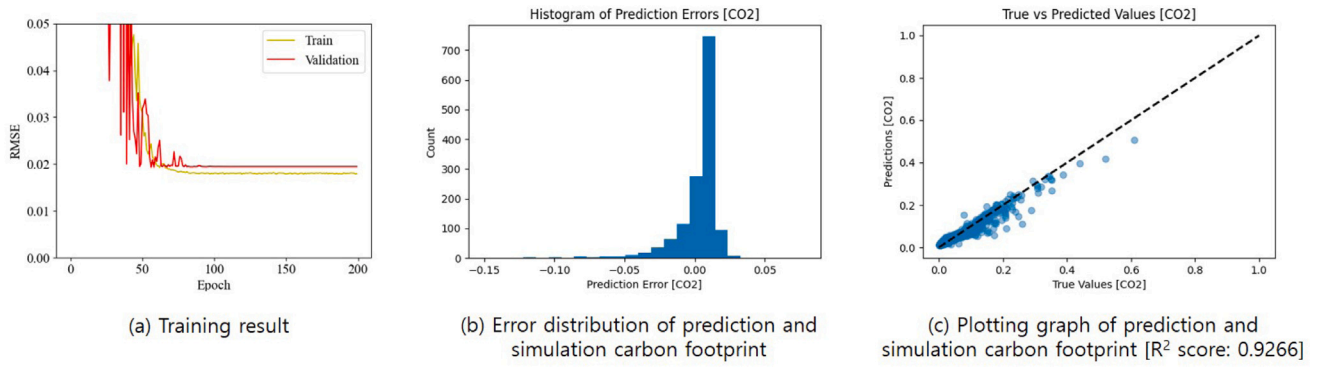


Fig. 8. Train and test results.

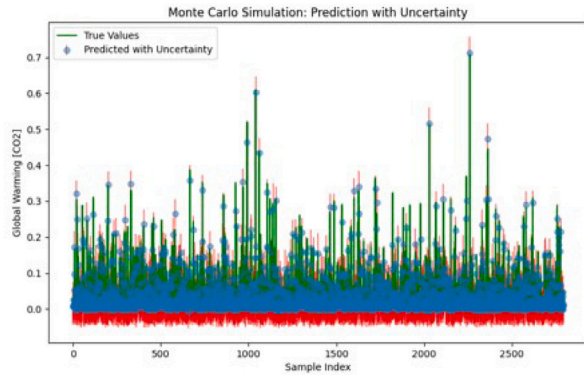


Fig. 9. Monte Carlo Simulation results for CO2 predictions. The plot compares true values (green line) against predicted values with associated uncertainties (blue points and red error bars) across samples, illustrating uncertainty quantification.

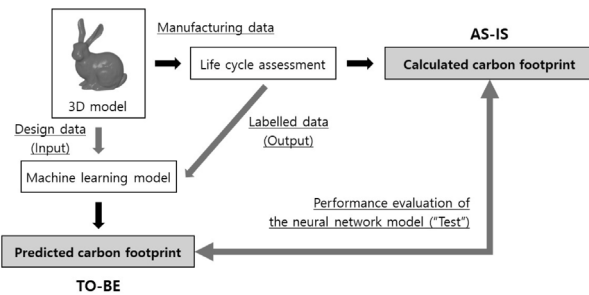


Fig. 10. Definition of predicted and calculated carbon footprint.

the true values from the test dataset, the blue dots indicate the mean predictions derived from 1000 Monte Carlo simulations, and the red vertical lines show the uncertainty bounds based on the standard deviation of the predictions. Most of the samples exhibit short red vertical lines, indicating low uncertainty (within $\pm 2\%$) in the model's predictions. This suggests that the predictive model is robust to small variations in the input data, such as $\pm 5\%$ changes in model volume or support volume. Additionally, the majority of the blue dots (mean predictions) closely align with the green line (true values), highlighting the model's overall accuracy. However, for samples with higher carbon footprints, the red vertical lines become longer, reflecting greater uncertainty (exceeding $\pm 5\%$). This tendency is observed primarily in large-volume models, where the sensitivity of the model to input variables like model volume is more pronounced. These results emphasize the model's ability to capture the complexities of manufacturing processes while also identifying areas where additional data or fine-tuning could enhance predictive reliability. Overall, the analysis demonstrates

that the predictive model performs robustly under varying conditions, with low uncertainty for most samples and reasonable reliability even in cases with higher variability.

As shown in Fig. 10, the carbon footprint was previously obtained by performing a Cradle to Gate LCA using additional manufacturing data. In this chapter, we proposed a method for predicting carbon footprint using an ANN-based predictive model trained on geometric data from 3D models in the design phase for simplified LCA. This predictive model is directly used in the optimization process to calculate fitness.

5. Grid-based part decomposition optimization

We propose the application of PD within the DfAM framework to enhance sustainability. For this purpose, a modular decomposition approach is adopted, generating both common and custom modules. Common modules are reusable across different products, while custom modules are specific to each model. By reusing common modules, it is possible to reduce material and energy consumption. In this study, grid-based decomposition is employed to create these modules. However, the results of decomposition can vary significantly depending on the position and angle of the grid, making it necessary to use an optimization algorithm to achieve the best decomposition results. The following subsection proposes a GA-based method to optimize the grid-based decomposition of 3D models.

5.1. PD optimization based on GA

GA is a metaheuristic algorithm based on the principles of natural selection and genetics, and it is effectively used to find optimal solutions for various problems. As illustrated in Fig. 11, GA operates through the following basic steps:

(1) Initialization: In this stage, an initial population of possible solutions is randomly generated. The population size is set to 30, and each individual (solution) is represented as a chromosome containing the center coordinates (x, y, z) and rotation angles (u, v, w) needed to position and rotate the 3D model within the grid. These chromosomes determine how the model will be decomposed into grid cells, which ultimately defines the resulting common and custom modules. Each generated solution is then evaluated for its fitness to identify which solution provides the most sustainable decomposition result.

(2) Fitness evaluation: In this step, each solution generated in the initialization phase is evaluated for fitness. The fitness evaluation determines how well the generated decomposition pattern achieves the sustainability objectives. In this study, three objective functions ($F1, F2, F3$) are used to evaluate fitness:

$F1$ aims to maximize the number of common modules, which are reusable across different products. By increasing the number of reusable

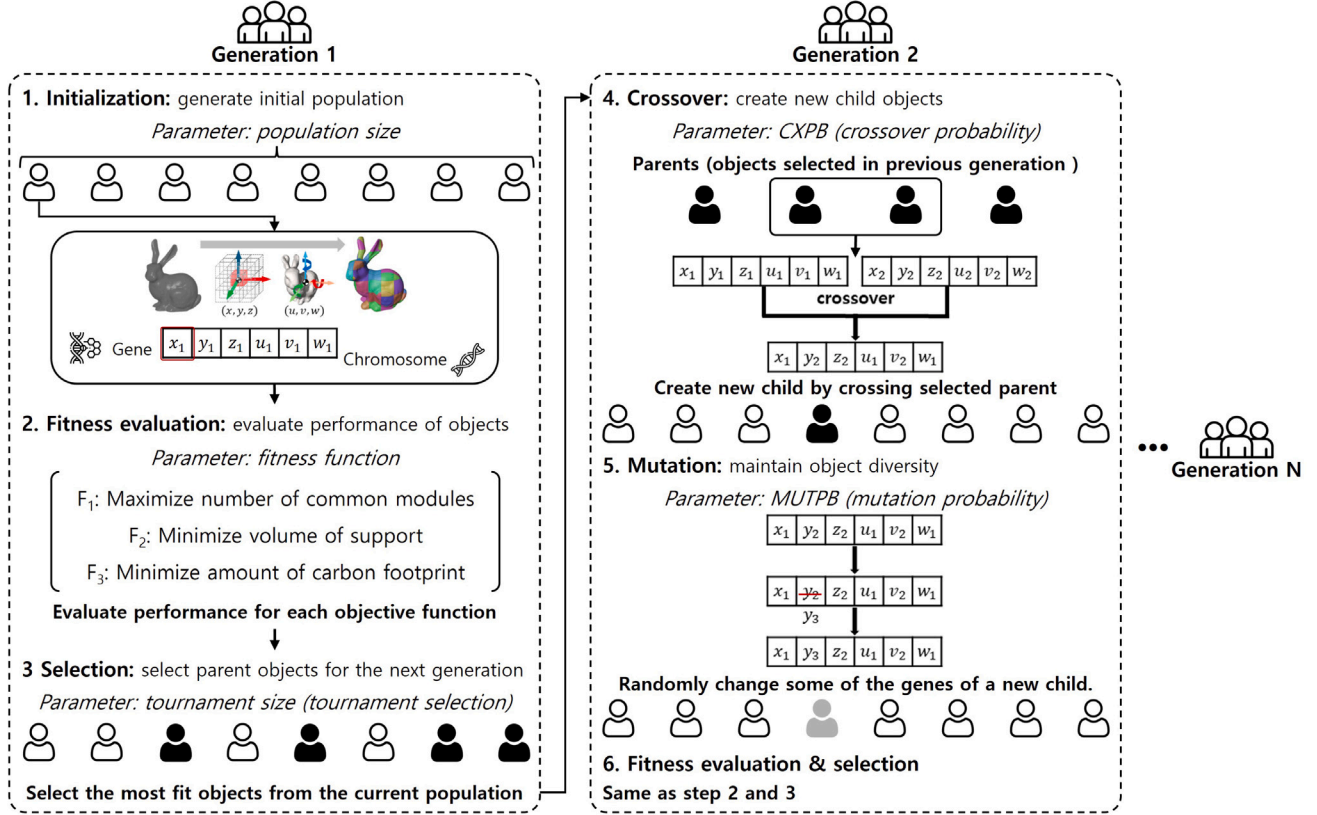


Fig. 11. Genetic algorithm process for optimizing grid-based decomposition.

modules, this function seeks to reduce material and energy consumption. By maximizing the number of modules, the sum of their volumes naturally increases as well. F1 is defined as follows:

$$F_1 = \sum_{i=1}^N \delta(M_i) \quad (3)$$

Here, N is the total number of modules, and $\delta(M_i)$ is a function that returns 1 if M_i is a common module, and 0 otherwise. This objective function determines the fitness based on the total number of common modules. The more common modules there are, the higher the fitness. Solutions optimized with F1 focus on maximizing the use of common modules, ultimately contributing indirectly to reducing the overall carbon footprint during the manufacturing process.

F2 aims to minimize the volume of support structure required during the manufacturing process for custom modules. Since common modules are reused, the actual manufacturing involves only the custom modules. Therefore, F2 targets the reduction of support structure volume specifically for custom modules, thereby enhancing manufacturing efficiency. F2 is defined as follows:

$$F_2 = \sum_{i=1}^{N_{\text{custom}}} V_{\text{support}}(M_i) \quad (4)$$

Here, N_{custom} represents the number of custom modules, and $V_{\text{support}}(M_i)$ is the support volume for the i th custom module. This objective function calculates fitness based on the total volume of support structure generated for custom modules. The smaller the support volume, the higher the fitness. Solutions optimized with F2 aim to minimize the support structure, reducing material usage and ultimately contributing to a lower carbon footprint in the overall manufacturing process.

F3 aims to minimize the carbon footprint generated during the manufacturing of custom modules, excluding common modules. Since common modules are reused, the carbon footprint is only generated from the production of custom modules. Therefore, F3 focuses on

minimizing the carbon footprint specifically for custom modules. F3 is defined as follows:

$$F_3 = \sum_{i=1}^{N_{\text{custom}}} C(M_i) \quad (5)$$

Here, N_{custom} represents the number of custom modules, and $C(M_i)$ is the carbon footprint associated with the i th custom module. This objective function determines fitness based on the total carbon footprint generated during the manufacturing of custom modules. The smaller the carbon footprint, the higher the fitness. F3 has the direct goal of minimizing the carbon footprint.

Each solution is evaluated based on the decomposition results of the 3D model, which depend on the central coordinates and rotation angles of the model, producing both common and custom modules. The fitness is calculated individually for each objective function. It is particularly important to evaluate the carbon footprint for solutions optimized using F1 and F2. Therefore, in the final assessment, the impact of the F1 and F2 results on the carbon footprint is analyzed and compared with the F3 results.

(3) Selection: After fitness evaluation, this step selects parent solutions for the next generation based on their fitness. The selection process uses a tournament selection method to ensure that solutions with higher fitness have a higher probability of being selected as parents. In this method, a subset of solutions is randomly chosen, and the one with the highest fitness within that subset is selected as a parent. This approach increases the likelihood of passing on good genes to the next generation while maintaining genetic diversity.

(4) Crossover: The selected parent solutions undergo cross-over to produce offspring solutions. During crossover, the genetic information of two parent solutions is combined to create new solutions. In this study, the crossover probability (CXPB) is set to 0.7. This probability determines how the genes from the parents is mixed into the offspring, allowing for exploration of various solutions. A lower crossover

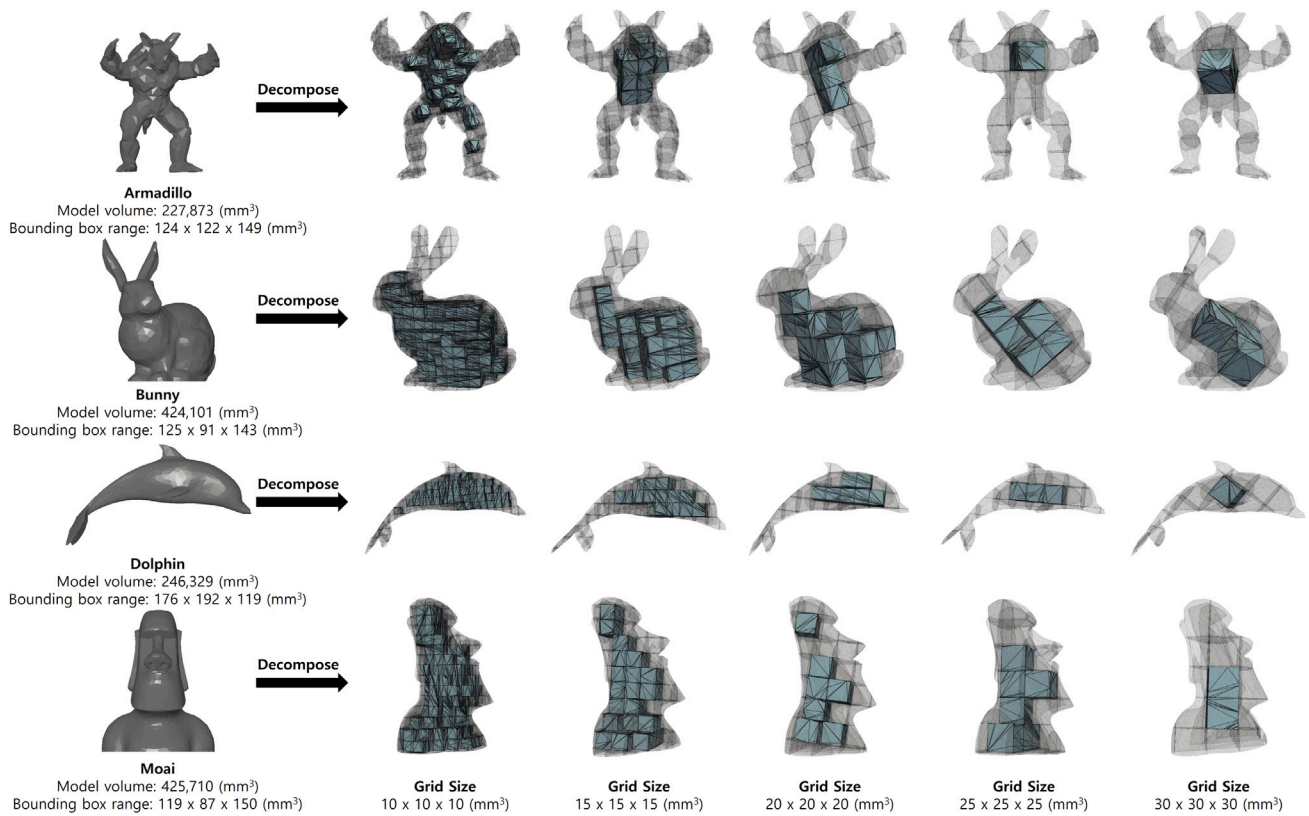


Fig. 12. Decomposition examples of test models as grid size increases.

probability might lead to insufficient genetic diversity, while a higher probability might disrupt good gene combinations too quickly.

(5) Mutation: The offspring solutions generated through crossover are then subjected to mutation. In this step, parts of the offspring's genetic information are randomly altered. The mutation probability (MUTPB) is set to 0.02, indicating how often a gene in the offspring will be changed. Mutation helps explore a wider solution space and prevents the algorithm from getting stuck in local optima. However, if the mutation rate is too low, the search might be too limited, and if too high, it might resemble random search, losing good solutions.

(6) Fitness evaluation and selection: In this stage, the fitness of the new population is evaluated, and the next generation is formed. The fitness of the newly generated offspring is assessed, and the current population is replaced by the new offspring, while the best solution from the current population is recorded. These processes are repeated according to the number of generations. In this study, the number of generations is set to 100 for optimization.

During the optimization process, various operations like shape analysis, transformation, and computation (e.g., decomposition) of 3D mesh models are necessary. In this study, the mesh processing library Trimesh⁶ was utilized for this purpose. Additionally, the volume of support structure for each module can vary greatly depending on orientation. Therefore, during the optimization process, it is essential to determine the optimal orientation for each module and calculate the corresponding support volume. The open-source tool Tweaker [58] was used to automatically find the optimal orientation that minimizes support structure for AM. Once the optimal orientation was determined, the model generate vertical supports for overhang surfaces below the critical angle, as shown in Fig. 5. PyVista⁷ was used for visualizing

and validating the decomposition results. PyVista is a library for 3D data visualization and analysis, helping to visually verify the optimized decomposition results and the orientation of each module.

In this study, the convergence of fitness values was monitored to determine whether the optimization had reached a satisfactory solution, with the algorithm terminating if the change in fitness between successive generations fell within a tolerance threshold. If the difference between the best solution in the current generation and the previous one remained within the tolerance for more than 20 consecutive generations, the algorithm was terminated. The DEAP (Distributed Evolutionary Algorithms in Python)⁸ framework was used for implementing the optimization algorithm.

5.2. Optimization results

We hypothesized that directly minimizing the carbon footprint during the manufacturing of custom modules would be more effective than simply increasing the number of common modules or minimizing support structures. This section compares and analyzes the optimization results for the three objective functions (F1, F2, F3) to validate our hypothesis.

The experiments were conducted using the four models shown in Fig. 12. The models were simplified using MeshLab to ensure efficient computation during the optimization process. Additionally, the bounding box range was set similarly across models to facilitate a meaningful comparison of the decomposition effects. The grid size for decomposition was varied in five increments (10, 15, 20, 25, 30) to compare the results based on the module size. While the grid and model sizes were chosen arbitrarily for the experiments, it is the relative size between model and grid that holds significance. The figure illustrate how the

⁶ <https://trimesh.org/>.

⁷ <https://docs.pyvista.org/>.

⁸ <https://github.com/deap/>.

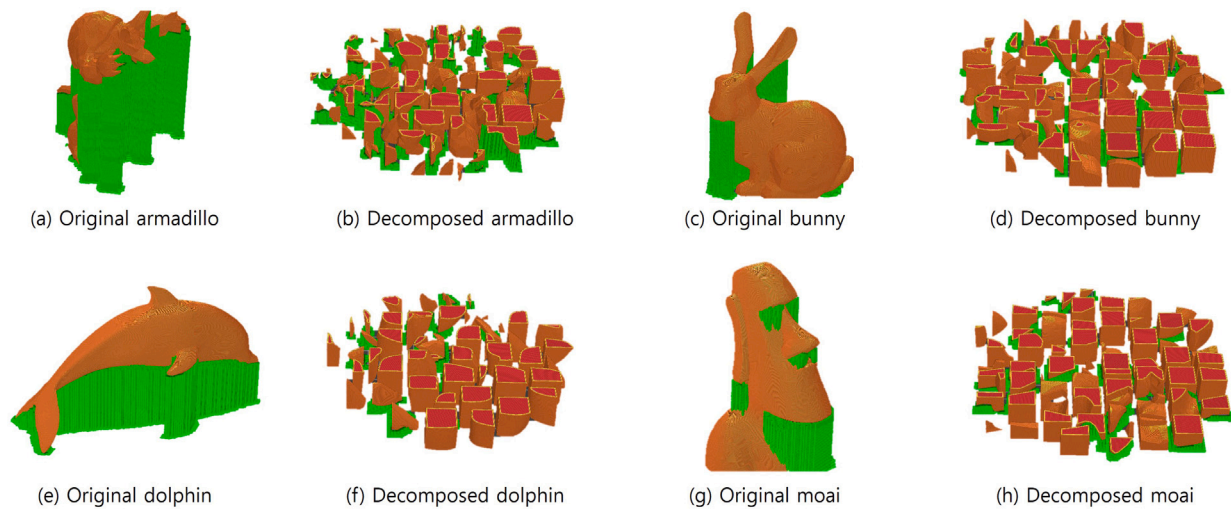


Fig. 13. Slicing results for both original and decomposed test models.

number of common modules changes with grid size, highlighting the substantial impact of grid placement and rotation.

Fig. 13 provides a visual representation of the benefits of decomposition by showing the slicing results before and after decomposition (excluding common modules). This allows for a visual assessment of the reduction in support structures and build height. While the benefits of PD might be less pronounced if a model can be produced in its entirety on a single AM device, this study assumes scenarios where the model is too large for the build space of the AM device, making decomposition necessary.

Fig. 14 shows the trends in the number of modules generated after optimal decomposition, and Table 3 provides the specific values. Fig. 14 normalizes the values for each grid size to the maximum value to facilitate comparison of the trends for each objective function. Subfigure (a) shows the trend in the total number of modules generated after decomposition. In FDM, the number of modules produced can increase the layer area on the build platform, potentially extending nozzle travel distance and increasing manufacturing time. Therefore, this can indirectly influence sustainability. Subfigure (b) shows the trend in the number of reusable common modules generated after decomposition. Since F1 maximizes the number of common modules, F1 consistently produced the highest values across all cases. Subfigure (c) shows the trend in the number of custom modules generated after excluding common modules. Subfigure (d) shows the trend in the number of small modules (with a volume less than 10% of the common module) generated. When optimized with F3, the total number of modules was minimized, which was closely related to the reduction in the number of small modules. Smaller modules are generally undesirable from a manufacturability and assembly perspective.

Fig. 15 illustrates the trends in support volume and carbon footprint generated after decomposition, and Table 4 provides the specific values. Fig. 15 also normalizes the values for each grid size to the maximum to facilitate comparison of the trends across the objective functions. Subfigure (a) shows the trend in the volume of support structure generated during module production. From a sustainability perspective, it is advantageous to minimize the support structure to save material and energy. As expected, optimization with F2 resulted in the smallest support volume. However, as shown in subfigures (b) and (c), minimizing the support volume did not necessarily translate into the lowest carbon footprint. Subfigure (b) shows the carbon footprint trend including common modules, while subfigure (c) excludes them. In most cases, optimizing with F3 naturally resulted in the lowest carbon footprint, which aligns with the hypothesis established when defining the F3 objective function. To minimize the carbon footprint, various

factors can play a role, such as increasing the number of common modules, reducing the number of small modules to shorten manufacturing time, or reducing the support structure volume to save material. These factors can be mutually exclusive, meaning that directly controlling the carbon footprint through F3 may be more effective than focusing on individual factors like F1 or F2. However, there were some exceptions, likely related to the characteristics of the optimization problem itself. Optimization algorithms, particularly metaheuristic methods, aim for global optima but may not always guarantee absolute optimal solutions due to the complexity and non-linearity of the problem. These algorithms utilize randomness and diversity in search paths to explore various solutions, and in complex problems, there may be many local optima, leading to unexpected results. For instance, F1 optimization could produce a lower carbon footprint in specific scenarios compared to F3 optimization, resulting in exceptional situations. Analyzing cases where exceptions occur can provide insights into the limitations and performance of the optimization algorithm used.

As shown in Table 4, optimizing with F3 resulted in the lowest carbon footprint for custom modules in most cases. This outcome is expected, as F3 was explicitly designed to minimize the carbon footprint during the manufacturing of custom modules. F1 optimization also produced relatively low carbon footprints, generally ranked second, which suggests that maximizing the number of common modules can indirectly contribute to reducing the carbon footprint by decreasing the total number of custom modules. However, the armadillo model, when decomposed with a grid size of 10, presented an exception where F1 slightly outperformed F3 in reducing the carbon footprint. This result can be attributed to the complexity of the armadillo model's geometry, which led to the generation of numerous small modules, complicating the optimization process and reducing the effectiveness of F3. Nevertheless, F1 managed to maintain a higher number of common modules, contributing to a marginally better carbon footprint.

These findings underscore the potential of F1 to positively impact sustainability by promoting the reuse of modules, and in certain cases, even outperforming F3. However, F2 optimization proved less effective in reducing the carbon footprint compared to F1 or F3, highlighting that merely minimizing support structure does not necessarily guarantee a lower carbon footprint, as other factors such as manufacturing time and energy consumption also play significant roles.

In conclusion, F3 demonstrated its effectiveness as a comprehensive approach to minimizing the carbon footprint by considering multiple factors during LCA. However, F1 also showed significant potential by enhancing module reusability, which contributed to meaningful carbon footprint reduction.

Table 3
The number of modules after decomposition.

Trend	Objective function	Armadillo					Bunny					Dolphin					Moai				
		Grid size					Grid size					Grid size					Grid size				
		10	15	20	25	30	10	15	20	25	30	10	15	20	25	30	10	15	20	25	30
Total modules	F1	568	244	124	78	59	768	282	154	93	68	536	214	116	83	59	785	289	150	95	56
	F2	579	244	143	87	65	741	283	147	95	66	543	215	113	74	50	774	298	142	91	54
	F3	565	204	107	78	50	700	264	136	83	51	508	183	98	53	41	699	273	123	72	52
Common modules	F1	73	13	4	1	1	241	51	18	6	3	112	22	7	3	1	223	51	15	7	2
	F2	66	12	2	0	0	224	44	13	4	0	95	14	2	1	0	187	47	8	1	0
	F3	65	10	4	1	0	215	48	16	6	3	108	18	4	0	0	198	50	10	3	2
Custom modules	F1	495	231	120	77	58	527	231	136	87	65	424	192	109	80	58	562	238	135	88	54
	F2	513	232	141	86	65	517	239	134	91	66	448	201	111	74	50	587	251	134	90	54
	F3	500	194	103	77	50	485	216	120	77	48	400	165	94	53	41	501	223	113	69	52
Small modules	F1	202	122	61	45	39	198	81	55	39	38	171	79	55	49	40	225	77	49	43	22
	F2	215	111	81	46	40	158	85	54	47	36	176	84	53	38	31	194	116	54	39	15
	F3	201	69	51	45	27	144	77	45	36	18	130	55	37	19	22	121	80	29	17	18

Table 4
Support volume and carbon footprint after decomposition.

Trend	Objective function	Armadillo					Bunny					Dolphin					Moai				
		Grid size					Grid size					Grid size					Grid size				
		10	15	20	25	30	10	15	20	25	30	10	15	20	25	30	10	15	20	25	30
Support volume	F1	339	565	766	1975	2636	34	136	199	424	520	236	418	552	716	449	2.7	16	9.3	84	122
	F2	127	273	576	1029	1422	20	92	114	182	189	12	28	50	72	123	1.7	0.7	0.2	5.4	2.4
	F3	272	616	1608	1975	2482	50	214	390	563	1612	218	392	566	660	340	2.5	8.2	8.0	64	84
Carbon footprint (total modules)	F1	166	86	54	41	35	237	111	75	56	47	159	78	52	42	34	241	114	74	56	44
	F2	167	86	59	43	36	230	112	73	57	47	161	79	51	40	32	239	116	71	55	44
	F3	164	77	50	41	32	220	107	70	52	42	153	72	47	35	29	221	110	66	48	43
Carbon footprint (custom modules)	F1	140	79	51	39	33	152	85	61	49	42	119	67	47	38	33	162	87	63	49	40
	F2	144	80	58	42	36	151	89	63	52	47	127	72	50	40	32	173	91	65	54	44
	F3	141	72	47	39	32	144	82	58	46	37	115	62	44	35	29	151	84	59	46	39

To further validate the observed trends in carbon footprint reduction, t-tests were conducted to assess the statistical significance of differences between the objective functions. The p-values summarized in Table 5 indicate that F3 optimization consistently leads to statistically significant improvements in carbon footprint reduction compared to F1 and F2 ($p < 0.05$). These results highlight the robustness of F3 in achieving sustainability goals across various grid sizes and models.

To evaluate the practical feasibility of the proposed genetic algorithm-based decomposition method, we conducted additional computational time analyses considering model geometry and grid size, as both factors significantly influence the complexity of decomposition. First, the computational time was analyzed based on the geometry of four models, each tested with a grid size of 20. The time per generation was calculated by dividing the total computation time by the number of generations required for convergence. The results showed that model geometry, rather than volume alone, has a significant impact on computation time. For instance, the Armadillo model exhibited the shortest time per generation (308.68 s) due to its simpler shape and fewer decomposition cases, despite its moderate volume (227,873 mm³). On the other hand, the Dolphin model had the longest time per generation (403.61 s), even though its volume was smaller (246,329 mm³), because of its higher shape complexity and greater number of potential decomposition cases. Second, the grid size was varied for the Dolphin model to assess its effect on computational time. As the grid size decreased, the number of potential decomposition cases increased exponentially, leading to longer computational times. For a grid size of 10, the time per generation was the longest at 2411.73 s, while a grid size of 30 reduced this time significantly to 159.12 s. This trend underscores the importance of grid size in managing computational efficiency during decomposition. These findings provide practical insights into the computational overhead of the proposed framework

Table 5
Summary of t-test results for carbon footprint and module count comparisons.

Comparison	Carbon footprint (p -value)	Module count (p -value)
F1 vs. F3	1.66×10^{-5}	5.71×10^{-5}
F2 vs. F3	8.94×10^{-7}	4.68×10^{-6}

and will guide future improvements to balance decomposition quality and efficiency. The detailed results are summarized in Table 6

The key findings of this experiment can be summarized as follows:

- When optimizing with the objective of minimizing the carbon footprint (F3), the decomposition resulted in fewer total modules, which was largely influenced by the reduction in the number of small modules. In FDM, an increase in the total number of modules typically extends the printing time per layer, leading to higher energy consumption, which significantly impacts the carbon footprint.
- When optimizing with the objective of minimizing support structures (F2), the support volume generated after decomposition was naturally minimized; however, the carbon footprint was higher compared to F1 and F3. This indicates that merely reducing material usage is not sufficient to achieve the lowest carbon footprint.
- In all cases where optimization was performed with the objective of minimizing the carbon footprint (F3), the expected result of the lowest carbon footprint during manufacturing was achieved. Although fewer common modules were secured compared to F1, it was confirmed that a sufficiently large number of common modules could still be obtained.

5.3. Other considerations

Users can leverage the proposed method to manufacture and assemble modules that minimize the carbon footprint after decomposition. While the experiments in this study assumed a 100% infill ratio

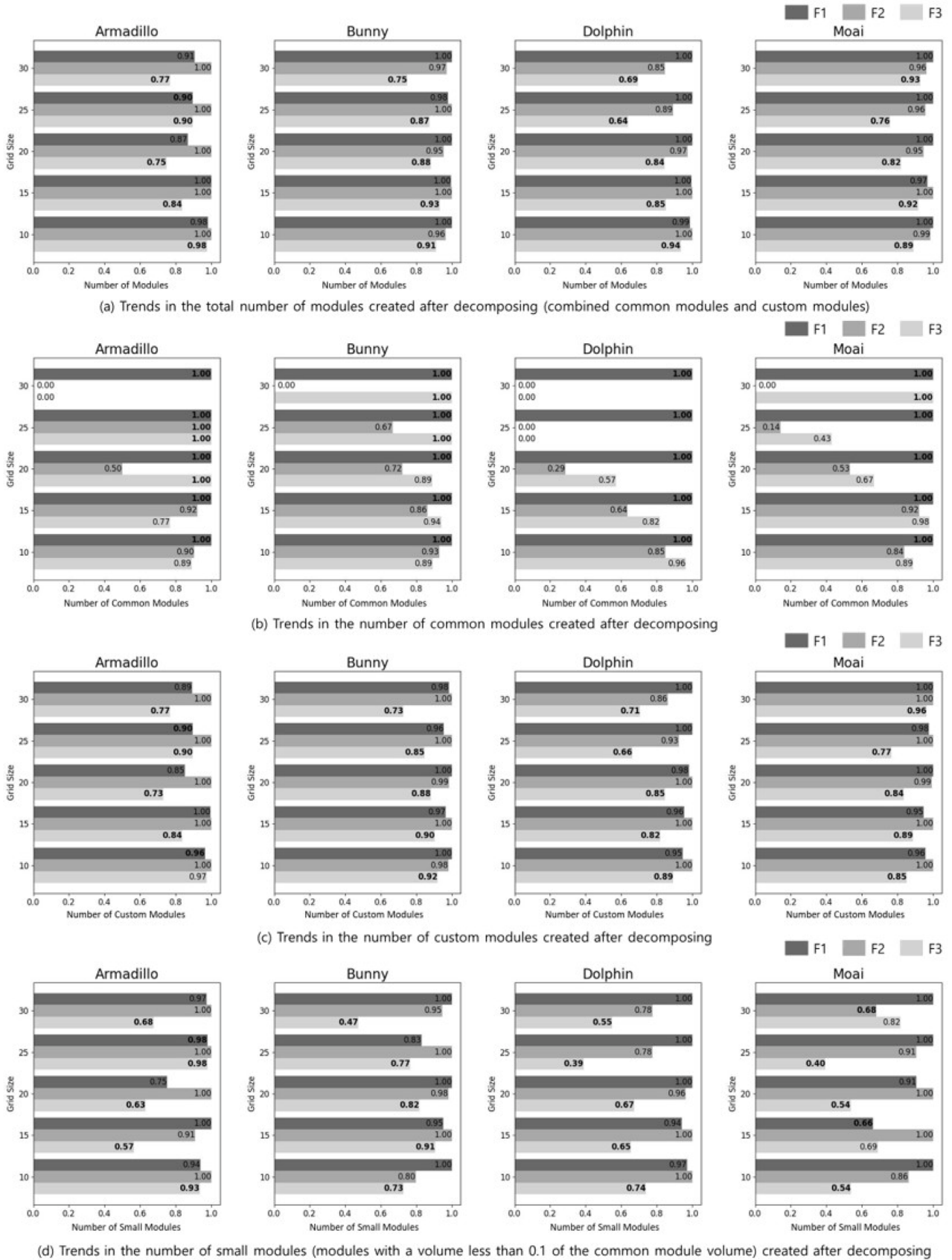


Fig. 14. Trends of the number of modules after decomposition. For each grid size, all values were normalized to the maximum.

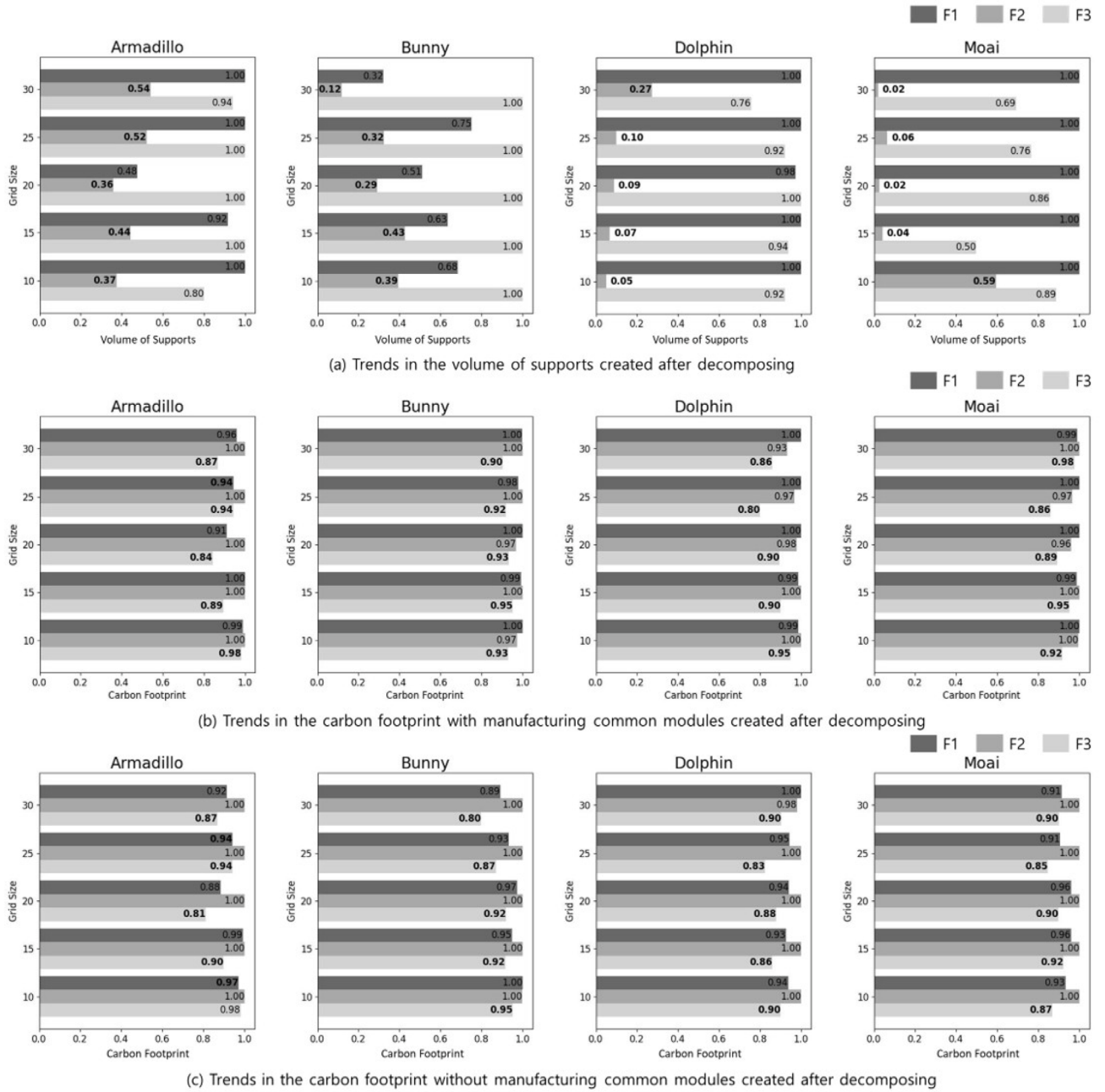


Fig. 15. Trends of support volume and carbon footprint after decomposition. For each grid size, all values were normalized to the maximum.

Table 6

Computational time per generation of GA by model and grid size. Results for grid size are experimental results of the dolphin model.

Parameter	Value	Time per generation (s)
Model geometry	Armadillo	308.68
	Bunny	342.89
	Dolphin	403.61
	Moai	332.30
Grid size	10	2,411.73
	15	775.97
	20	403.61
	25	217.14
	30	159.12

Table 7

Comparison of printing time, material usage, and carbon footprint for two infill ratios after decomposition.

Model	Infill ratio (%)	Printing time	Material usage (g)	Carbon footprint (g)
armadillo	100	2 d 20 h 53 m	239	34.0
	20	1 d 21 h 41 m	139	22.5
bunny	100	3 d 20 h 50 m	357	46.0
	20	2 d 7 h 32 m	187	27.4
dolphin	100	2 d 20 h 30 m	262	33.9
	20	1 d 16 h 1 m	134	19.7
moai	100	4 d 7 h 19 m	411	51.3
	20	2 d 12 h 27 m	211	29.9

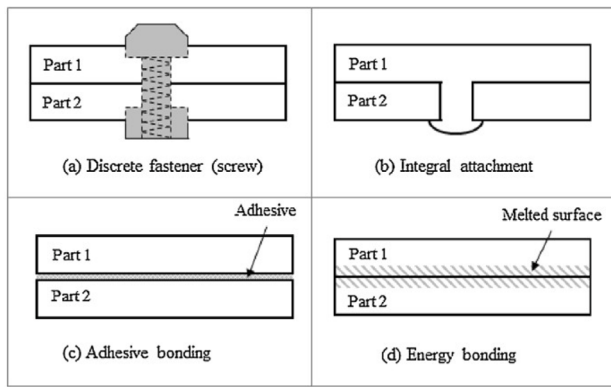


Fig. 16. Four connection types: (a) discrete fastener (screw), (b) integral attachment, (c) adhesive bonding, and (d) energy bonding [34].

for simplicity, in practice, lower infill ratios are often used. Table 7 demonstrates how reducing the infill ratio to 20% can further decrease manufacturing time, material usage, and the carbon footprint.

The grid-based decomposition method was chosen to facilitate the acquisition and reuse of common modules. However, in cases where the grid size is large or the model is thin, common modules may not be generated. Even in such scenarios, the benefits of carbon footprint minimization-based optimization, as proposed in this study, remain significant. Additionally, PD is always followed by assembly stage, making the choice of appropriate interfaces and connectors crucial. Since the interface can affect the decomposition results, some studies also considered assemblability for optimization [3,42]. As shown in Fig. 16, common assembly methods for modules include discrete fastening (e.g., screws), integral attachment, adhesive bonding, and energy bonding [34]. These methods can be selected based on the structural and functional requirements of the assembled product. Incorporating such considerations ensures the practical applicability of the proposed method in real-world manufacturing scenarios.

To maximize the benefits of PD, modules can be manufactured using multiple AM devices, thereby shortening the manufacturing time per device [56,62,63]. In this case, while energy consumption per device may decrease due to shorter build time, additional energy consumption from using multiple devices must be considered. Therefore, a thorough comparison between the total energy consumption of using multiple devices versus a single device should be conducted. If the carbon footprint is not significantly impacted, the reduction in manufacturing time from using multiple devices could offer substantial cost and time saving.

Considering other printing methods besides FDM may offer additional advantages. In FDM, increased layer area prolongs nozzle travel distance and manufacturing time. In contrast, Digital Light Processing (DLP) method hardens each layer simultaneously using a projector, meaning that layer area has little impact on manufacturing time. Thus, in DLP methods, smaller module sizes would reduce time, although DLP machines typically have more limited build sizes compared to FDM. Therefore, it is crucial to select the appropriate printing method based on the product requirements, build size, and decomposition objectives.

6. Future work and limitations

This study proposes a framework that combines grid-based part decomposition and machine learning-based LCA to minimize the carbon footprint of AM. The framework aims to enhance sustainability during the early design stages, which inherently introduced certain limitations to the study's scope. While these limitations were intentionally set to ensure clarity and feasibility during the early design phase, they may constrain the broader applicability of some results. The following

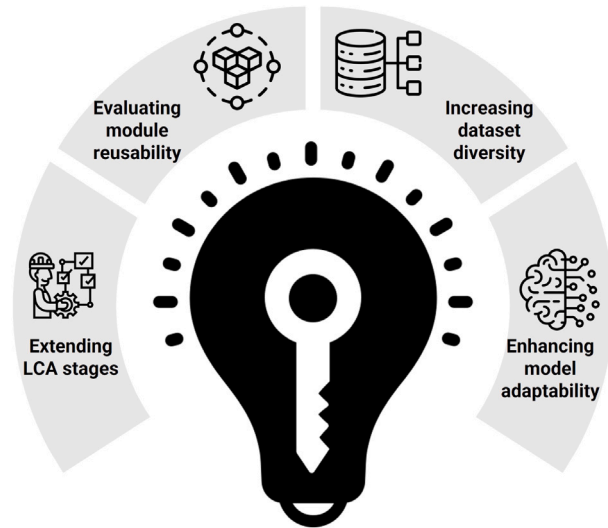


Fig. 17. Future research directions: extending LCA stages, evaluating module reusability, increasing dataset diversity, enhancing model adaptability.

sections summarize the key limitations and discuss future research directions to address them (see Fig. 17).

Firstly, this study focuses on the manufacturing stage through a Cradle-to-Gate assessment, excluding assembly and post-processing phases. This was an intentional decision to ensure clarity and feasibility during the early design phase, effectively analyzing the carbon footprint of manufacturing processes and achieving module decomposition optimization. Since assembly and post-processing methods vary significantly depending on product type, material, and application, generalizing their environmental impact within a unified LCA framework presents considerable challenges. As a result, this study prioritizes the manufacturing phase, where key parameters influencing carbon footprint, such as material usage, printing time, and energy consumption, can be more reliably quantified. However, the volume and number of modules can impact assembly complexity and resource efficiency, making assembly and post-processing critical factors in the practical application of module designs. Future research should explore methods to systematically incorporate these stages into the LCA framework, potentially by developing a structured database of energy consumption and emissions for various post-processing and assembly techniques.

Secondly, this study focuses on enhancing sustainability during the early design phase through the reuse of common modules but does not quantitatively incorporate their long-term effects on carbon footprint reduction. The current F3 (carbon footprint minimization) objective emphasizes minimizing the direct carbon footprint of the manufacturing process, leaving the potential resource savings and carbon reductions from repeated reuse of common modules as a future research avenue. For instance, reusing common modules beyond a certain number of cycles or employing injection molding instead of FDM can enhance production efficiency and economic viability. However, limitations such as material defects in high-durability components and constraints from economic or physical damage could restrict reuse feasibility. D-EComposer focuses on a decomposition strategy centered on sustainability and manufacturability, and the evaluation of optimal design strategies for functional or structural parts is beyond the scope of this research and will be considered in another future research. Future research should focus on quantitatively analyzing the long-term effects of module reuse on carbon footprint reduction and assessing its practical feasibility.

Thirdly, the Baumann dataset was utilized as a foundational resource for AM carbon footprint prediction but may not fully capture the specific characteristics of industrial components or topology-optimized

design elements. While the dataset, sourced from Thingiverse, underwent filtering to exclude large files and elongated models for processing efficiency, this approach reduced dataset diversity and limited the framework's generalizability to certain complex geometries. Additionally, the LCA prediction model demonstrated sensitivity to input variables in specific scenarios. Monte Carlo simulation revealed prediction uncertainties exceeding $\pm 5\%$ for large-volume models, highlighting the need to enhance data quality and model training for improved reliability. In this study, D-ECOMposer was experimentally validated using the FDM process; however, this does not imply that it is limited to a specific AM process. Since the proposed methodology predicts the carbon footprint based on geometric features rather than relying on the unique characteristics of a particular AM process, it can be applied to other processes such as LPBF and SLS through dataset development following the same principles. Future research should focus on developing datasets that encompass greater geometric diversity and complexity, alongside improving the model's robustness to enhance the generalizability of the framework.

Fourthly, the ANN-based predictive model has limitations in adaptability to changes in manufacturing parameters, such as print speed and device power. Print speed affects manufacturing time, power consumption, and carbon emissions, while variations in device power alter energy usage over a fixed duration. Since this study's approach prioritizes rapid and efficient prediction, it relies on geometric information rather than directly incorporating detailed manufacturing parameters, which inherently limits the model's adaptability to parameter variations. To address this limitation, future research should explore ways to incorporate additional feature data, such as process-specific parameters, while maintaining the model's efficiency and generalizability within the optimization framework. Additionally, assessing the impact of parameter changes on LCA outcomes and improving the model's adaptability through further data collection and retraining will be key to enhancing its flexibility and reliability.

The limitations outlined in this study are closely tied to the scope defined to ensure clarity and feasibility during the early design stage. Despite these limitations, this research provides a practical framework for supporting sustainability-driven design decisions and contributes significantly to the effective evaluation of carbon footprints in manufacturing processes during the design phase. Furthermore, the identified limitations can be addressed in future research, allowing for the expansion of the framework into a more comprehensive and practical sustainability assessment. This advancement would support balanced and holistic decision-making during the early design stage, ultimately maximizing sustainability in manufacturing processes.

7. Conclusion

In recent years, there has been extensive research on DfAM to maximize the advantages of AM. While previous studies have explored the benefits of PC from a sustainability perspective, few have considered the advantages of PD from the same perspective. This study hypothesizes that PD has significant potential for enhancing sustainability due to its modular design strategies. By leveraging common modules, PD can reduce material waste and promote recyclability, aligning with design-for-sustainability principles. These advantages are particularly pronounced when decomposition strategies are applied in a way that maximizes resource efficiency. While the benefits of PD might be less pronounced if a model can be produced in its entirety on a single AM device, this study assumes scenarios where the model exceeds the build size of the AM device, making decomposition necessary.

D-ECOMposer adopts a modular PD approach that utilizes common modules to maximize the potential for reuse, thereby enhancing sustainability. A key contribution of this study is the development of a data-driven carbon footprint prediction model that allows for quick and accurate assessments of environmental impact during the early design stage. By using this model as an objective function in the PD

optimization process, the study presents a method for minimizing the carbon footprint through optimal PD.

The primary contribution of D-ECOMposer is the introduction of a novel approach to maximize the sustainability of PD in AM. This framework provides a solution that enables the prediction of the carbon footprint with minimal information during the early stages of design, thereby deriving the optimal decomposition strategy. Compared to traditional empirical methods, it facilitates quick and accurate decision-making, serving as a crucial tool for sustainable manufacturing.

From a practical application standpoint, D-ECOMposer can significantly benefit large-scale manufacturers and companies engaged in custom production using AM by addressing the challenge of managing multiple modules for large-scale parts. In scenarios where the size of the parts exceeds the build capacity of AM devices, PD becomes indispensable. The framework proposes strategies to optimize module design for efficient assembly and minimizes post-processing efforts, thereby enhancing overall sustainability. It allows these companies to make decisions that minimize environmental impact right from the design stage, offering substantial long-term advantages in terms of cost saving and regulatory compliance. Specifically, by maximizing the use of recyclable common modules, D-ECOMposer can enhance resource efficiency.

While this study focused on minimizing the carbon footprint in AM, real-world manufacturing environments also require consideration of factors such as manufacturing time, cost, and assemblability. Therefore, future research will aim to further develop D-ECOMposer by incorporating these additional factors through a multi-objective optimization approach. Specifically, exploring methods to integrate F1, F2, and F3 into a balanced and aggregated objective function will be a key focus. This will allow for a more comprehensive framework that supports early-stage design decisions while considering the trade-offs between performance, cost, and sustainability. Additionally, small custom modules generated through decomposition may pose manufacturing challenges, and in some cases, merging certain custom modules could yield better sustainability outcomes. Conversely, further subdivision of certain modules may also be beneficial depending on the design and manufacturing constraints. Hence, further optimization will be conducted to appropriately merge or further subdivide adjacent custom modules, excluding common modules. Lastly, future plans include developing a method to overlay carbon footprint prediction results onto a 3D model using color, enabling designers to intuitively make sustainable design decisions [64].

CRedit authorship contribution statement

Minseok Ko: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yeongjun Yoon:** Validation, Software, Methodology. **Jaeyeon Kim:** Software, Data curation. **Samyeon Kim:** Writing – review & editing, Writing – original draft, Investigation. **Soonjo Kwon:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] M. Gebler, A.J.S. Uiterkamp, C. Visser, A global sustainability perspective on 3D printing technologies, *Energy Policy* 74 (2014) 158–167.
- [2] T. Kellner, How 3D Printing Will Change Manufacturing, GE Reports, 2017.
- [3] Y. Oh, H. Ko, T. Sprock, W.Z. Bernstein, S. Kwon, Part decomposition and evaluation based on standard design guidelines for additive manufacturability and assemblability, *Addit. Manuf.* 37 (2021) 101702.
- [4] X. Jiang, X. Cheng, Q. Peng, L. Liang, N. Dai, M. Wei, C. Cheng, Models partition for 3D printing objects using skeleton, *Rapid Prototyp. J.* 23 (1) (2017) 54–64.
- [5] Y. Oh, C. Zhou, S. Behdad, Part decomposition and 2D batch placement in single-machine additive manufacturing systems, *J. Manuf. Syst.* 48 (2018) 131–139.
- [6] W.M. Wang, C. Zanni, L. Kobbelt, Improved surface quality in 3D printing by optimizing the printing direction, in: *Computer Graphics Forum*, Vol. 35, Wiley Online Library, 2016, pp. 59–70.
- [7] J. Gregory, S. Fredholm, R. Kirchain, Is economic value an effective proxy for embodied energy and environmental impact in material systems? in: 2009 IEEE International Symposium on Sustainable Systems and Technology, IEEE, 2009, pp. 1–5.
- [8] ISO 14001, Environmental Management Systems — Requirements with Guidance for Use, ISO, 2015.
- [9] ISO 14044, Environmental Management — Life Cycle Assessment — Requirements and Guidelines, ISO, 2006.
- [10] I. 14040, Environmental Management — Life Cycle Assessment — Principles and Framework, ISO, 2006.
- [11] P.C. Priarone, G. Ingarao, Towards criteria for sustainable process selection: On the modelling of pure subtractive versus additive/subtractive integrated manufacturing approaches, *J. Clean. Prod.* 144 (2017) 57–68.
- [12] R. Huang, M. Riddle, D. Graziano, J. Warren, S. Das, S. Nimbalkar, J. Cresko, E. Masanet, Energy and emissions saving potential of additive manufacturing: the case of lightweight aircraft components, *J. Clean. Prod.* 135 (2016) 1559–1570.
- [13] G. Ingarao, P.C. Priarone, Y. Deng, D. Paraskevas, Environmental modelling of aluminium based components manufacturing routes: Additive manufacturing versus machining versus forming, *J. Clean. Prod.* 176 (2018) 261–275.
- [14] J. Faludi, Z. Hu, S. Alrashed, C. Braunholz, S. Kaul, L. Kassaye, Does material choice drive sustainability of 3D printing? *Int. J. Mech. Aerosp. Ind. Mechatronics Eng.* (2015).
- [15] Y. Li, B.S. Linke, H. Voet, B. Falk, R. Schmitt, M. Lam, Cost, sustainability and surface roughness quality—A comprehensive analysis of products made with personal 3D printers, *CIRP J. Manuf. Sci. Technol.* 16 (2017) 1–11.
- [16] E. Dalpadulo, F. Pini, F. Leali, A CO₂ emissions life cycle assessment of additive and conventional manufacturing based lightweight design in the automotive, in: *ASME International Mechanical Engineering Congress and Exposition*, Vol. 87592, American Society of Mechanical Engineers, 2023, V002T02A023.
- [17] R. Lyons, A. Newell, P. Ghadimi, N. Papakostas, Environmental impacts of conventional and additive manufacturing for the production of Ti-6Al-4V knee implant: a life cycle approach, *Int. J. Adv. Manuf. Technol.* 112 (2021) 787–801.
- [18] G.M. Cappucci, M. Pini, P. Neri, M. Marassi, E. Bassoli, A.M. Ferrari, Environmental sustainability of orthopedic devices produced with powder bed fusion, *J. Ind. Ecol.* 24 (2020) 681–694.
- [19] I. Dani, W.G. Drossel, N. Milaev, H. Korn, C. Hannemann, J. Hohlfeld, R. Wertheim, Sustainability of industrial components using additive manufacturing and foam materials, *Procedia Manuf.* 43 (2020) 10–17.
- [20] H. Hegab, N. Khanna, A. Monib, A. Salem, Design for sustainable additive manufacturing: A review, *Sustain. Mater. Technol.* 35 (2023) e00576.
- [21] H.-S. Yoon, J.-Y. Lee, H.-S. Kim, M.-S. Kim, E.-S. Kim, Y.-J. Shin, W.-S. Chu, S.-H. Ahn, A comparison of energy consumption in bulk forming, subtractive, and additive processes: Review and case study, *Int. J. Precis. Eng. Manuf.- Green Technol.* 1 (2014) 261–279.
- [22] J. Watson, K. Taminger, A decision-support model for selecting additive manufacturing versus subtractive manufacturing based on energy consumption, *J. Clean. Prod.* 176 (2018) 1316–1322.
- [23] S. Kilchert, K. Hoshcke, A. Pfaff, K. Kappe, M. May, Towards more sustainable aviation: Life cycle analysis of a topology optimized 3D printed cargo door latch fitting, in: *AIAA AVIATION FORUM ASCEND 2024*, 2024.
- [24] K. Hoshcke, Sustainable design with topology optimization for laser powder bed fusion of metals, *Fraunhofer-Inst. Kurzzeitdynamik Ernst- Mach- Inst. EMI* (2021).
- [25] K. Hoshcke, K. Kappe, S. Patil, S. Kilchert, J. Kim, A. Pfaff, Sustainability-oriented topology optimization towards a more holistic design for additive manufacturing, in: *International Conference on Additive Manufacturing in Products and Applications*, 2023.
- [26] F. Iraldo, F. Testa, I. Bartolozzi, An application of life cycle assessment (LCA) as a green marketing tool for agricultural products: the case of extra-virgin olive oil in Val di Cornia, Italy, *J. Environ. Plan. Manag.* 57 (1) (2014) 78–103.
- [27] N. Rajagopalan, M.M. Bilec, A.E. Landis, Life cycle assessment evaluation of green product labeling systems for residential construction, *Int. J. Life Cycle Assess.* 17 (2012) 753–763.
- [28] P. Karka, S. Papadokonstantakis, A. Kokossis, Digitizing sustainable process development: From ex-post to ex-ante LCA using machine-learning to evaluate bio-based process technologies ahead of detailed design, *Chem. Eng. Sci.* 250 (2022) 117339.
- [29] S. Agatonovic-Kustrin, R. Beresford, Basic concepts of artificial neural network (ANN) modeling and its application in pharmaceutical research, *J. Pharm. Biomed. Anal.* 22 (5) (2000) 717–727.
- [30] A. D'Amico, G. Ciulla, M. Traverso, V.L. Brano, E. Palumbo, Artificial neural networks to assess energy and environmental performance of buildings: An Italian case study, *J. Clean. Prod.* 239 (2019) 117993.
- [31] P. Hou, O. Jolliet, J. Zhu, M. Xu, Estimate ecotoxicity characterization factors for chemicals in life cycle assessment using machine learning models, *Environ. Int.* 135 (2020) 105393.
- [32] A.Z. Naser, F. Defersha, E. Pei, Y.F. Zhao, S. Yang, Toward automated life cycle assessment for additive manufacturing: A systematic review of influential parameters and framework design, *Sustain. Prod. Consum.* (2023).
- [33] A.Z. Naser, F. Defersha, X. Xu, S. Yang, Automating life cycle assessment for additive manufacturing with machine learning: Framework design, dataset buildup, and a case study, *J. Manuf. Syst.* 71 (2023) 504–526.
- [34] Y. Oh, C. Zhou, S. Behdad, Part decomposition and assembly-based (Re) design for additive manufacturing: A review, *Addit. Manuf.* 22 (2018) 230–242.
- [35] S. Yang, Y. Tang, Y.F. Zhao, A new part consolidation method to embrace the design freedom of additive manufacturing, *J. Manuf. Process.* 20 (2015) 444–449.
- [36] S. Yang, W. Min, J. Ghibardo, Y.F. Zhao, Understanding the sustainability potential of part consolidation design supported by additive manufacturing, *J. Clean. Prod.* 232 (2019) 722–738.
- [37] D. Son, S. Kim, B. Jeong, Sustainable part consolidation model for customized products in closed-loop supply chain with additive manufacturing hub, *Addit. Manuf.* 37 (2021) 101643.
- [38] A.L.R. Prathyusha, G.R. Babu, A review on additive manufacturing and topology optimization process for weight reduction studies in various industrial applications, *Mater. Today: Proc.* 62 (2022) 109–117.
- [39] D.J. Munk, J.D. Miller, Topology optimization of aircraft components for increased sustainability, *AIAA J.* 60 (2022) 445–460.
- [40] F. Borda, A.D. La Rosa, L. Filice, F. Gagliardi, Environmental impact of process constrained topology optimization design on automotive component life, *Int. J. Mater. Form.* 16 (2023) 48.
- [41] A. Kumar, D. Chhabra, Parametric topology optimization approach for sustainable development of customized orthotic appliances using additive manufacturing, *Mech. Adv. Mater. Struct.* 31 (2024) 5276–5289.
- [42] L. Luo, I. Baran, S. Rusinkiewicz, W. Matusik, Chopper: Partitioning models into 3D-printable parts, *ACM Trans. Graph.* 31 (6) (2012) 1–9.
- [43] M. Yao, Z. Chen, L. Luo, R. Wang, H. Wang, Level-set-based partitioning and packing optimization of a printable model, *ACM Trans. Graph.* 34 (6) (2015) 1–11.
- [44] İ. Demir, D.G. Aliaga, B. Benes, Near-convex decomposition and layering for efficient 3D printing, *Addit. Manuf.* 21 (2018) 383–394.
- [45] X. Chen, H. Zhang, J. Lin, R. Hu, L. Lu, Q.X. Huang, B. Chen, Dapper: decompose-and-pack for 3d printing, *ACM Trans. Graph.* (2015).
- [46] X. Chen, H. Li, C.-W. Fu, H. Zhang, D. Cohen-Or, B. Chen, 3D fabrication with universal building blocks and pyramidal shells, *ACM Trans. Graph.* 37 (6) (2018) 1–15.
- [47] S. Kim, S.K. Moon, Sustainable product family configuration based on a platform strategy, *J. Eng. Des.* 28 (10–12) (2017) 731–764.
- [48] G. Zou, Z. Li, C. He, A Sustainable Product Multi-Platform Planning Model for Assembly and Disassembly Process, *J. Mech. Des.* (ISSN: 1050-0472) 146 (9) (2024) 092001, <http://dx.doi.org/10.1115/1.4064675>.
- [49] S. Kim, S.K. Moon, Eco-modular product architecture identification and assessment for product recovery, *J. Intell. Manuf.* 30 (2019) 383–403.
- [50] M. Kwak, H. Kim, Green profit maximization through integrated pricing and production planning for a line of new and remanufactured products, *J. Clean. Prod.* (ISSN: 0959-6526) 142 (2017) 3454–3470, <http://dx.doi.org/10.1016/j.jclepro.2016.10.121>.
- [51] M. INOUE, S. YAMADA, S. MIYAJIMA, K. ISHII, R. HASEBE, K. AOYAMA, T. YAMADA, S. BRACKE, A modular design strategy considering sustainability and supplier selection, *J. Adv. Mech. Des. Syst. Manuf.* 14 (2) (2020) JAMDSM0023, <http://dx.doi.org/10.1299/jamdsm.2020jamdsm0023>.
- [52] F. De Fazio, C. Bakker, B. Flipsen, R. Balkenende, The Disassembly Map: A new method to enhance design for product reparability, *J. Clean. Prod.* (ISSN: 0959-6526) 320 (2021) 128552, <http://dx.doi.org/10.1016/j.jclepro.2021.128552>.
- [53] S. Kim, S.K. Moon, Sustainable platform identification for product family design, *J. Clean. Prod.* (ISSN: 0959-6526) 143 (2017) 567–581.

- [54] S. Kim, S.K. Moon, A part consolidation design method for additive manufacturing based on product disassembly complexity, *Appl. Sci.* (ISSN: 2076-3417) 10 (3) (2020) <http://dx.doi.org/10.3390/app10031100>.
- [55] N. Chiriac, K. Hölttä-Otto, D. Lysy, E. Suk Suh, Level of Modularity and Different Levels of System Granularity, *J. Mech. Des.* (ISSN: 1050-0472) 133 (10) (2011) 101007, <http://dx.doi.org/10.1115/1.4005069>.
- [56] L. Choi, M. Choi, S. Kwon, D. Youn, G. Song, Modular production of small ship models using 3D printing for model tests, *Ocean Eng.* 302 (2024) 117685.
- [57] F.W. Baumann, Thirty thousand 3D models from thingiverse, *Data* 3 (1) (2018) 5.
- [58] C. Schranz, Tweaker-auto rotation module for FDM 3D printing, *Salzbg. Res. Salzbg. Austria* (2016).
- [59] G. Wernet, C. Bauer, B. Steubing, et al., The ecoinvent database version 3 (part I): overview and methodology, *Int. J. Life Cycle Assess.* 21 (2016) 1218–1230, <http://dx.doi.org/10.1007/s11367-016-1087-8>.
- [60] C. Caceres-Mendoza, P. Santander-Tapia, F.A.C. Sanchez, N. Troussier, M. Camargo, H. Boudaoud, Life cycle assessment of filament production in distributed plastic recycling via additive manufacturing, *Clean. Waste Syst.* 5 (2023) 100100.
- [61] D.P. Kingma, J. Ba, Adam: A method for stochastic optimization, 2014, arXiv preprint [arXiv:1412.6980](https://arxiv.org/abs/1412.6980).
- [62] Y. Oh, P. Witherell, Y. Lu, T. Sprock, Nesting and scheduling problems for additive manufacturing: A taxonomy and review, *Addit. Manuf.* 36 (2020) 101492.
- [63] Y. Oh, Y. Cho, Scheduling of build and post processes for decomposed parts in additive manufacturing, *Addit. Manuf.* 59 (2022) 103164.
- [64] T. Vernica, M. Glišić, B. Veluri, D. Ramanujan, Volumetric mapping and visualization of life cycle assessment results on computer-aided design models, *J. Clean. Prod.* 418 (2023) 138035.